

Problem-Solving Methods

REFERENCE SHEET

Methods, key formulas, worked examples and pictures for the problem types of a first university mathematics course, in four parts: Part I a selector for integrals and quick-reference tables, Part II Calculus, Part III Algebra, trigonometry, geometry and discrete mathematics, and Part IV Multivariable and vector calculus. Each block gives a *method* (the moves, in the order you should try them), the *key formulas*, one or two *worked examples* carried to the end, the *traps* that cost marks, and a picture wherever a picture does the thinking for you. It ends with 88 practice problems and their answers.

The first thirty seconds of any problem

- 1. Name the object.** Integral, derivative, limit, equation, count, locus, proof? The blocks below are grouped by exactly that classification.
- 2. Look for the structure before the calculation.** A derivative sitting next to its function, a symmetric interval, a hidden identity, a factorisation, an invariant. Most “hard” problems on the sheet are one observation followed by routine work.
- 3. Draw it.** A 10-second sketch of the region, the curve, the triangle or the vector configuration settles the limits, the sign and the orientation, which is where most marks are lost.
- 4. Estimate the answer before you find it:** its sign, its size, its units. A closed form that disagrees with a rough estimate is wrong.
- 5. Check at the end** by the cheapest independent route: differentiate an antiderivative, substitute a solution back, test a special case, or compare with the estimate.

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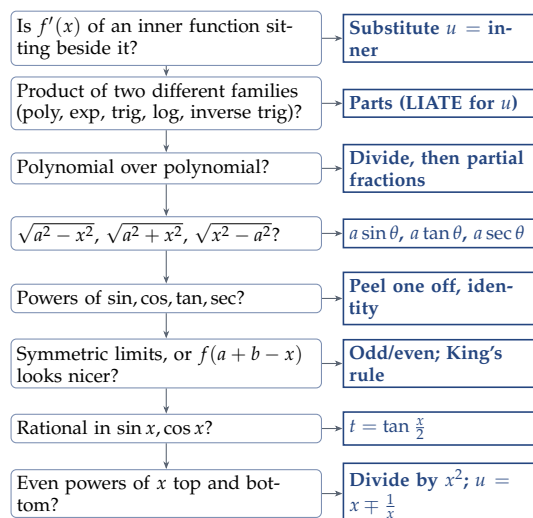
Part I · Integral selector

BLOCKS I1–I2

Integral technique selector

I1–I4

Method: ask in this order Simplify first (expand, split the fraction, use an identity). Then go down the chart ($\downarrow$ = no, $\rightarrow$ = yes) and take the *first* yes. If two moves both apply, the one that removes the ugliest piece wins.



Key formulas

$$\int_a^b f(x) dx = \int_a^b f(a + b - x) dx \quad (\text{King's rule})$$

$$\int_{-a}^a f = \begin{cases} 0 & f \text{ odd} \\ 2 \int_0^a f & f \text{ even} \end{cases}$$

$$t = \tan \frac{x}{2} : \sin x = \frac{2t}{1+t^2}, \cos x = \frac{1-t^2}{1+t^2}, dx = \frac{2dt}{1+t^2}$$

$$\int_0^{\pi/2} \sin^n x dx = \int_0^{\pi/2} \cos^n x dx = \frac{n-1}{n} I_{n-2}$$

Example King's rule. Evaluate $I = \int_0^{\pi/2} \frac{\sin^5 x}{\sin^5 x + \cos^5 x} dx$.

Solution Replace x by $\frac{\pi}{2} - x$: sine and cosine swap, so $I = \int_0^{\pi/2} \frac{\cos^5 x}{\sin^5 x + \cos^5 x} dx$. Add the two forms: the integrand becomes 1, so $2I = \frac{\pi}{2}$.

$\Rightarrow I = \pi/4$ (the exponent 5 is irrelevant).

Example Divide by x^2 . Evaluate $\int_0^\infty \frac{x^2 + 1}{x^4 + 1} dx$.

Solution Divide top and bottom by x^2 : $\frac{1 + x^{-2}}{x^2 + x^{-2}}$. With $u = x - \frac{1}{x}$, $du = (1 + x^{-2}) dx$ and $x^2 + x^{-2} = u^2 + 2$; as x runs from 0 to ∞ , u runs from $-\infty$ to ∞ :

$$\int_{-\infty}^\infty \frac{du}{u^2 + 2} = \frac{1}{\sqrt{2}} \left[\arctan \frac{u}{\sqrt{2}} \right]_{-\infty}^\infty.$$

$$\Rightarrow \pi/\sqrt{2}.$$

Traps Change the limits when you substitute, or convert back; never both. After a trigonometric substitution check the sign of $\sqrt{\cos^2 \theta}$ on the range. A “clever” symmetry only works if the interval is fixed by the map.

Derivatives, equations, limits and logic

D, DE, L, P

Derivatives

$f(x)^{g(x)}$	take logs: $\frac{y'}{y} = g' \ln f + g \frac{f'}{f}$
$F(x, y) = 0$	differentiate both sides, collect y'
$x(t), y(t)$	$\frac{dy}{dx} = \frac{\dot{y}}{\dot{x}}, \frac{d^2y}{dx^2} = \frac{\frac{d}{dt}(dy/dx)}{\dot{x}}$
inverse $g = f^{-1}$	$g'(b) = \frac{1}{f'(a)}$ where $f(a) = b$
$\int_{u(x)}^{v(x)} h$	$h(v)v' - h(u)u'$

First-order ODEs

$y' = f(x)g(y)$	separate: $\int \frac{dy}{g} = \int f dx$
$y' + Py = Q$	$\mu = e^{\int P}$, then $(\mu y)' = \mu Q$
$P' = rP(1 - \frac{P}{K})$	$P = \frac{K}{1 + Ae^{-rt}}$; fastest at $P = \frac{K}{2}$
$T' = -k(T - T_a)$	$T = T_a + (T_0 - T_a)e^{-kt}$

Limits

Try in order: substitute; factor or use a conjugate; a standard limit; a Maclaurin series; l'Hopital (only for $\frac{0}{0}$ or $\frac{\infty}{\infty}$); for $1^\infty, 0^0, \infty^0$ take logarithms first.

$$\frac{\sin x}{x}, \frac{\tan x}{x}, \frac{e^x - 1}{x}, \frac{\ln(1+x)}{x} \rightarrow 1, \quad \frac{1 - \cos x}{x^2} \rightarrow \frac{1}{2}.$$

Example Find $\lim_{x \rightarrow \infty} \left(1 + \frac{2}{x}\right)^{3x}$.

Solution $\ln y = 3x \ln(1 + \frac{2}{x}) = 3x(\frac{2}{x} - \frac{2}{x^2} + \dots) \rightarrow 6.$
 $\Rightarrow e^6.$

Example Find $\lim_{x \rightarrow 0} \frac{x - \sin x}{x^3}$.

Solution $\sin x = x - \frac{x^3}{6} + O(x^5)$, so the numerator is $\frac{x^3}{6} + O(x^5).$
 $\Rightarrow \frac{1}{6}.$

Logic puzzles

Invariant	a quantity no move changes: parity, sum, product, sum of squares
Colouring	chessboard colours count what a tile can cover
Pigeonhole	$n + 1$ objects in n boxes: two share a box
Extremal	look at the largest, smallest or first element
Information	k weighings give at most 3^k outcomes
Small cases	solve $n = 1, 2, 3, 4$, guess, then prove

Example Five points lie in a unit square. Show two are at most $\frac{\sqrt{2}}{2}$ apart.

Solution Cut the square into four $\frac{1}{2} \times \frac{1}{2}$ squares. Five points, four squares: two points share one, and the diagonal of a small square is $\frac{\sqrt{2}}{2}$.

Part II · Calculus

BLOCKS 1–21

Limits, continuity, IVT and MVT

1

Method *Continuity at a* : the two one-sided limits exist, agree, and equal $f(a)$. *Existence of a root*: continuity plus a sign change (IVT). *Exactly one root*: IVT for existence, then strict monotonicity (or Rolle) for uniqueness. *An inequality*: apply the MVT on $[a, b]$ and bound $f'(c)$.

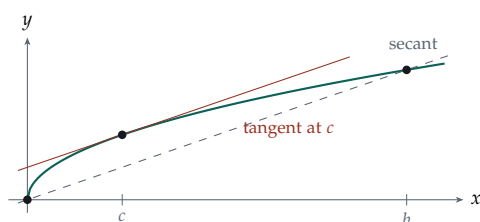
Problem type → first move

$\frac{0}{0}$ rational limit	factor and cancel
difference of roots	multiply by the conjugate
limit at $\pm\infty$ with roots	divide by the top power; $\sqrt{x^2} = x $
piecewise and continuous	equate both one-sided limits to $f(a)$
exactly one root	IVT for existence, monotonic for uniqueness
an inequality	MVT on the interval, bound f'
oscillating factor	squeeze between $\pm$ bounds

Key formulas

$$\text{MVT: } \frac{f(b) - f(a)}{b - a} = f'(c) \text{ for some } c \in (a, b)$$

$$\text{Squeeze: } g \leq f \leq h, \quad g, h \rightarrow L \Rightarrow f \rightarrow L$$



MVT: some tangent is parallel to the chord. For $\sqrt{x}$ on $[0, 4]$, $c = 1$.

Example Show that $10 + \frac{1}{22} < \sqrt{101} < 10 + \frac{1}{20}$.

Solution MVT for $\sqrt{x}$ on $[100, 101]$: $\sqrt{101} - 10 = \frac{1}{2\sqrt{c}}$ with $100 < c < 101$. So $\frac{1}{2\sqrt{101}} < \sqrt{101} - 10 < \frac{1}{20}$, and $\sqrt{101} < 11$ gives the left bound.

⇒ both inequalities, with no calculator.

Harder example Show that $x^3 + x - 1 = 0$ has exactly one real root, and locate it in an interval of length $\frac{1}{4}$.

Solution $f(0) = -1 < 0 < 1 = f(1)$, so IVT gives a root in $(0, 1)$; $f'(x) = 3x^2 + 1 > 0$, so f is strictly increasing and the root is unique. $f(0.5) = -0.375$ and $f(0.75) = 0.171875$.

⇒ the only root lies in $(0.5, 0.75)$ (it is $0.6823\dots$).

Traps The IVT needs continuity on the *closed* interval; $1/x$ changes sign on $[-1, 1]$ with no root. A piecewise function must be checked from both sides at every join.

Differentiation from first principles

2

Method Write $\frac{f(x+h) - f(x)}{h}$ and *simplify until h cancels* before letting $h \rightarrow 0$: expand polynomials; combine fractions over a common denominator; multiply roots by the conjugate; use the addition formulas and the standard limits for $\sin$, $\cos$, e^x , $\ln$.

Problem type → first move

polynomial	expand $(x+h)^n$, cancel h
reciprocal	common denominator
root	multiply by the conjugate
$\sin$, $\cos$	addition formula, standard limits
e^{kx} , $\ln x$	factor out, $\frac{e^h - 1}{h}$, $\frac{\ln(1+t)}{t}$
kink or join	two one-sided quotients

Key formulas

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$\frac{\sin h}{h} \rightarrow 1, \quad \frac{\cos h - 1}{h} \rightarrow 0$$

Example Differentiate $f(x) = \frac{1}{x^2}$ from first principles.

Solution

$$\frac{\frac{1}{(x+h)^2} - \frac{1}{x^2}}{h} = \frac{x^2 - (x+h)^2}{h x^2 (x+h)^2} = \frac{-(2x+h)}{x^2 (x+h)^2}$$

Now let $h \rightarrow 0$.

$$\Rightarrow f'(x) = -2/x^3.$$

Example Is $f(x) = |x|$ differentiable at 0?

Solution $\frac{|h|}{h} = +1$ for $h > 0$ and -1 for $h < 0$; the one-sided limits differ.

⇒ no: a corner.

Traps Letting $h \rightarrow 0$ before cancelling gives $\frac{0}{0}$. At a join or a kink, compute the two one-sided limits separately.

Chain, product, quotient, implicit

3

Method Read the function from the *outside in*: the last operation decides the rule. For an implicit curve, differentiate every term with respect to x , attaching y' to each y -term, then collect the y' terms and solve.

Problem type → first move

composite	chain rule, outside in
three factors	$(uvw)' = u'vw + uv'w + uvw'$
constant over a function	rewrite as a negative power
implicit curve	differentiate, collect y'
horizontal / vertical tangent	numerator / denominator of y' is 0
implicit y''	differentiate again, substitute y'

Key formulas

$$(uv)' = u'v + uv', \quad \left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$$

$$(f \circ g)' = f'(g) g'$$

Example Find the gradient of $x^2 + xy + y^3 = 11$ at $(1, 2)$.

Solution $2x + (y + xy') + 3y^2y' = 0$, so $y'(x + 3y^2) = -(2x + y)$. At $(1, 2)$: $y' \cdot 13 = -4$.

$$\Rightarrow y' = -\frac{4}{13}.$$

Second derivative of an implicit curve: differentiate the y' equation again, then substitute y' .

Harder example Find $\frac{d^2y}{dx^2}$ on the circle $x^2 + y^2 = 25$.

Solution $y' = -\frac{x}{y}$. Differentiate the quotient: $y'' = -\frac{y - xy'}{y^2} = -\frac{y + x^2/y}{y^2} = -\frac{x^2 + y^2}{y^3}$, and use the curve.

$\Rightarrow y'' = -25/y^3$: concave down on the upper half, up on the lower.

Traps The xy term needs the product rule, and its y carries a y' . A quotient is often easier as u/v .

Logarithmic differentiation and inverses

4

Method Variable base and variable exponent, or a long product or quotient: take $\ln$ first, differentiate, multiply back by y . Derivative of an inverse at a point: find a with $f(a) = b$, then $g'(b) = 1/f'(a)$; you never need g itself.

Problem type $\rightarrow$ first move

$x^{g(x)}$ or f^g	take $\ln$ of both sides
long product or quotient	$\ln$ turns it into a sum
$(f^{-1})'(b)$	solve $f(a) = b$, take $1/f'(a)$
inverse trig of a composite	table plus chain rule
prove an inverse derivative	differentiate $f(f^{-1}(x)) = x$

Key formulas

$$(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}, \quad (\arccos x)' = \frac{-1}{\sqrt{1-x^2}}$$

$$(\arctan x)' = \frac{1}{1+x^2}$$

$$(a^x)' = a^x \ln a, \quad (\log_a x)' = \frac{1}{x \ln a}$$

$$g'' = -\frac{f''}{(f')^3} \text{ for } g = f^{-1}$$

Example Differentiate $y = x^{\sqrt{x}}$ and evaluate at $x = 4$.

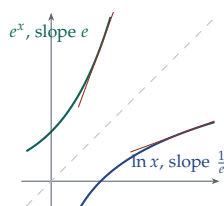
Solution $\ln y = \sqrt{x} \ln x$, so $\frac{y'}{y} = \frac{\ln x}{2\sqrt{x}} + \frac{1}{\sqrt{x}}$. At $x = 4$, $y = 16$ and $\frac{y'}{y} = \frac{\ln 4}{4} + \frac{1}{2}$.

$$\Rightarrow y'(4) = 8 + 8 \ln 2.$$

Example $f(x) = x^3 + 2x - 1$ and $g = f^{-1}$. Find $g'(2)$.

Solution $f(1) = 2$ and $f'(1) = 5$.

$$\Rightarrow g'(2) = \frac{1}{5}.$$



Reflect in $y = x$: slopes at matching points are reciprocals.

Traps $\frac{d}{dx} x^x$ is neither $x \cdot x^{x-1}$ nor $x^x \ln x$; it is the sum of the two.

Parametric and vector-valued curves

5

Method Gradient $\dot{y}/\dot{x}$. Horizontal tangent: $\dot{y} = 0 \neq \dot{x}$; vertical: $\dot{x} = 0 \neq \dot{y}$; both zero: a cusp, look closer. For $\frac{d^2y}{dx^2}$ differentiate $\frac{dy}{dx}$ in t and divide by $\dot{x}$. To find the Cartesian equation, eliminate t .

Problem type $\rightarrow$ first move

gradient at a parameter	$\dot{y}/\dot{x}$
horizontal / vertical tangent	$\dot{y} = 0 / \dot{x} = 0$
concavity	sign of d^2y/dx^2
self-intersection	$t_1 \neq t_2$ giving one point
Cartesian equation	eliminate t : $\cos^2 + \sin^2 = 1$, $(t + \frac{1}{t})^2 - (t - \frac{1}{t})^2 = 4$
speed, arc length	$ \mathbf{r}' $, $\int \mathbf{r}' dt$

Key formulas

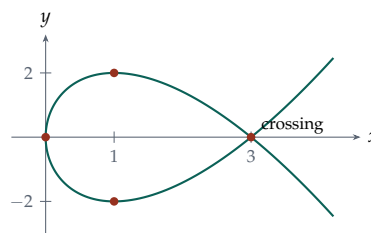
$$\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}(\frac{dy}{dx})}{\dot{x}}, \quad \text{speed } |\mathbf{r}'(t)| = \sqrt{\dot{x}^2 + \dot{y}^2}$$

Example For $x = t^2$, $y = t^3 - 3t$, find the horizontal and vertical tangents and $\frac{d^2y}{dx^2}$.

Solution $\frac{dy}{dx} = \frac{3t^2-3}{2t}$. Horizontal at $t = \pm 1$, i.e. $(1, \mp 2)$; vertical at $t = 0$, i.e. $(0, 0)$. Differentiating in t and dividing by $2t$:

$$\frac{d^2y}{dx^2} = \frac{3(t^2 + 1)}{4t^3}.$$

The curve crosses itself where $t = \pm\sqrt{3}$, at $(3, 0)$.



Harder example Find the length of one arch of the cycloid $x = t - \sin t$, $y = 1 - \cos t$, $0 \leq t \leq 2\pi$.

Solution $\dot{x}^2 + \dot{y}^2 = (1 - \cos t)^2 + \sin^2 t = 2 - 2\cos t = 4\sin^2 \frac{t}{2}$, so the speed is $2\sin \frac{t}{2} \geq 0$ on the range.

$$L = \int_0^{2\pi} 2\sin \frac{t}{2} dt = \left[-4\cos \frac{t}{2} \right]_0^{2\pi}.$$

$$\Rightarrow L = 8.$$

Traps $\frac{d^2y}{dx^2}$ is not $\ddot{y}/\ddot{x}$. Check $\dot{x} \neq 0$ before calling a point a horizontal tangent.

Tangents, normals and linear approximation

6

Method Tangent $y - y_0 = m(x - x_0)$, normal gradient $-1/m$. *Tangent through an outside point*: write the tangent at a general point a , make it pass through the point, solve for a . *Approximation*: $f(a + h) \approx f(a) + hf'(a)$; concave down means the tangent lies above, so the estimate is too big.

Problem type → first move

tangent / normal at a point	$m = f'(x_0)$, normal $-1/m$
tangent through an outside point	general tangent at a , impose the point
tangent parallel to a line	solve $f'(x) = m$
estimate a root or power	tangent at the nearest easy point
too big or too small?	concavity
measurement error	differentials; relative error

Key formulas

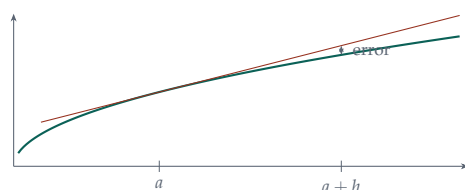
$$f(a + h) \approx f(a) + hf'(a)$$

$$\delta V \approx V'(r) \delta r, \quad \frac{\delta V}{V} \approx n \frac{\delta r}{r} \quad (V \propto r^n)$$

Example Estimate $\sqrt[3]{8.3}$ and say whether the estimate is high or low.

Solution $f = x^{1/3}$, $a = 8$, $f'(8) = \frac{1}{3 \cdot 4} = \frac{1}{12}$, so $f(8.3) \approx 2 + \frac{0.3}{12} = 2.025$. Since $f'' < 0$ the tangent lies above the curve.

⇒ 2.025, an overestimate (true value 2.02474...).



Harder example Find every tangent to $y = x^3$ that passes through $(2, 0)$.

Solution The tangent at $x = a$ is $y = a^3 + 3a^2(x - a)$. Put $(x, y) = (2, 0)$: $0 = a^3 + 3a^2(2 - a) = 2a^2(3 - a)$, so $a = 0$ or $a = 3$.

⇒ $y = 0$ and $y = 27x - 54$.

Traps Work in radians for any trigonometric approximation. Percentage error in r^n is n times the percentage error in r .

Optimisation

8

Method (1) Name the quantity to optimise and the variables. (2) Use the constraint to reduce to *one* variable. (3) State the domain. (4) Solve $f' = 0$. (5) *Justify* max or min: sign change of f' , sign of f'' , or compare the critical values with the endpoints.

Related rates

7

Method (1) Draw it and name everything that changes. (2) Write one equation linking the quantities: geometry, similar triangles, Pythagoras. (3) Differentiate *with respect to t*. (4) Only now substitute the numbers for the instant asked about.

Problem type → first move

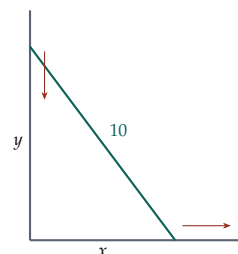
sphere, balloon	$V = \frac{4}{3}\pi r^3$, differentiate in t
conical tank	similar triangles remove r first
ladder, two ships	Pythagoras
shadow	similar triangles
angle of elevation	$\tan \theta = \frac{h}{x}$, then $\sec^2 \theta \dot{\theta}$
trough	cross-section area $\times$ length

Key formulas

$$V_{\text{sphere}} = \frac{4}{3}\pi r^3, \quad S_{\text{sphere}} = 4\pi r^2$$

$$V_{\text{cone}} = \frac{1}{3}\pi r^2 h, \quad V_{\text{cyl}} = \pi r^2 h$$

Example A 10 m ladder slides so that its foot moves away from the wall at 1 m/s. How fast is the top falling when the foot is 6 m out?



Solution $x^2 + y^2 = 100$, so $2x\dot{x} + 2y\dot{y} = 0$. At $x = 6$, $y = 8$: $\dot{y} = -\frac{6 \cdot 1}{8}$.

⇒ the top falls at 0.75 m/s.

Harder example A plane flies level at 3 km altitude directly away from an observer at 600 km/h. How fast is the angle of elevation changing when the horizontal distance is 4 km?

Solution $\tan \theta = \frac{3}{x}$, so $\sec^2 \theta \dot{\theta} = -\frac{3\dot{x}}{x^2}$. At $x = 4$: $\sec^2 \theta = 1 + \frac{9}{16} = \frac{25}{16}$, so $\dot{\theta} = \frac{-3 \cdot 600/16}{25/16} = -72$.

⇒ falling at 72 rad/h, that is 1.2 rad/min.

Traps Substituting $x = 6$ before differentiating turns the variable into a constant. Keep the sign: a falling height has negative $\dot{y}$.

Problem type → first move

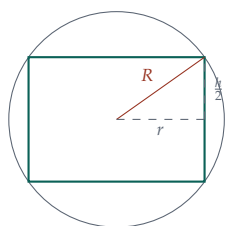
box or can	constraint removes one dimension
shape inscribed in a curve	put a vertex on the curve: $(x, f(x))$
nearest point on a curve	minimise d^2 , not d
fence along a wall	the wall side needs no fence
cone or cylinder in a sphere	Pythagoras on the cross-section
path through two media	$T = \frac{d_1}{v_1} + \frac{d_2}{v_2}$

Example A rectangular pen uses a wall as one side and 120 m of fence for the other three. Maximise the area.

Solution Two sides of length y , one of length $x = 120 - 2y$: $A = (120 - 2y)y$, $A' = 120 - 4y = 0$ at $y = 30$; $A'' = -4 < 0$.

$$\Rightarrow 60 \times 30 \text{ m, area } 1800 \text{ m}^2.$$

Harder example Find the cylinder of largest volume inside a sphere of radius R .



$$\text{Cross-section: } r^2 + \frac{h^2}{4} = R^2.$$

Solution $V = \pi r^2 h = \pi(R^2 - \frac{h^2}{4})h$, so $V' = \pi(R^2 - \frac{3h^2}{4}) = 0$ at $h = \frac{2R}{\sqrt{3}}$, a maximum because $V'' < 0$.

$$\Rightarrow h = \frac{2R}{\sqrt{3}}, V_{\max} = \frac{4\pi R^3}{3\sqrt{3}}.$$

Traps A critical point on an open interval is only a global maximum after a justification. Minimising d^2 is legitimate; minimising d gives the same point with worse algebra.

Curve sketching

9

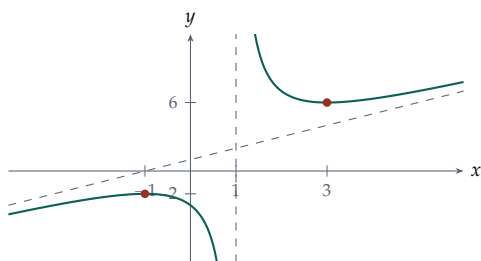
Method Domain; intercepts; symmetry (even/odd); asymptotes (vertical where the denominator vanishes and the numerator does not; horizontal or oblique by long division); sign chart of f' for turning points; sign chart of f'' for concavity and inflection. Then plot the key points and join them.

Problem type → first move

rational function	long division gives the oblique asymptote
$x^n e^{-x}$, $\frac{\ln x}{x}$	exponentials beat powers beat logs
constants from conditions	one equation per condition
stationary inflection	$f' = 0$ and f' does not change sign
cubic	inflection is the midpoint of the turning points

Example Sketch $y = \frac{x^2 + 3}{x - 1}$.

Solution Division: $y = x + 1 + \frac{4}{x-1}$, so the asymptotes are $x = 1$ and $y = x + 1$. $y' = 1 - \frac{4}{(x-1)^2} = 0$ at $x = -1$ (max, $y = -2$) and $x = 3$ (min, $y = 6$).



Harder example Find the stationary points and inflections of $y = x^4 - 4x^3$.

Solution $y' = 4x^2(x - 3)$: stationary at 0 and 3. The factor x^2 does not change sign, so $x = 0$ is a stationary inflection; $x = 3$ is a minimum, $y = -27$. $y'' = 12x(x - 2)$ changes sign at 0 and 2.

$\Rightarrow$ min $(3, -27)$; inflections at $(0, 0)$ and $(2, -16)$.

Traps A factor that cancels makes a hole, not an asymptote. $f'' = 0$ alone is not an inflection (x^4 at 0).

Kinematics

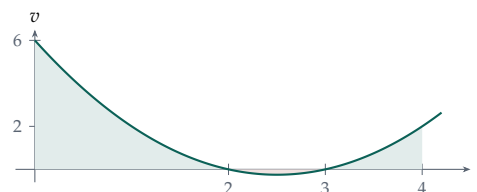
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Method $s \xrightarrow{d/dt} v \xrightarrow{d/dt} a$; go back by integrating and fixing each constant from an initial condition. The particle turns where v changes sign. Distance is $\int |v| dt$: split at every sign change. Speeding up when v and a have the same sign.

Problem type → first move

at rest / turning	$v = 0$ and sign change
given a , find s	integrate twice, two constants
distance travelled	$\int v $, split at the zeros of v
greatest height	$v = 0$
a depends on v	separate $\frac{dv}{dt} = a(v)$
a depends on s	$a = v \frac{dv}{ds}$

Example $v(t) = t^2 - 5t + 6$ on $0 \leq t \leq 4$. Find the displacement and the distance.



Areas below the axis count – for displacement, + for distance.

Solution $v = (t - 2)(t - 3)$. With $S(t) = \frac{t^3}{3} - \frac{5t^2}{2} + 6t$: $S(2) = \frac{14}{3}$, $S(3) = \frac{9}{2}$, $S(4) = \frac{16}{3}$.

$\Rightarrow$ displacement $\frac{16}{3}$; distance $\frac{14}{3} + \frac{1}{6} + \frac{5}{6} = \frac{17}{3}$.

Harder example With resistance proportional to speed, $\dot{v} = -kv$, $v(0) = v_0$. How far does the particle go?

Solution $v = v_0 e^{-kt}$, $s = \frac{v_0}{k} (1 - e^{-kt}) \rightarrow \frac{v_0}{k}$.

$\Rightarrow$ a finite distance v_0/k , never quite reached.

Traps Displacement is not distance; speed is $|v|$. Fix a positive direction and keep it.

Integration by substitution

11

Method Look for an inner function whose derivative is also present (up to a constant). Set u equal to it, replace dx , change the limits. Any leftover x must be written in terms of u ($x = u - 1$, $x^2 = u^2 - a$). Mixed roots: $x = u^k$ with k the lcm of the root orders.

Problem type → first move

f'/f	$\ln f $
$f' f^n$	$\frac{f^{n+1}}{n+1}$
$g' e^g$	e^g
$f'/(1+f^2)$	$\arctan f$
$f'/\sqrt{1-f^2}$	$\arcsin f$
polynomial $\times \sqrt{ax+b}$	$u = ax+b$ or $u = \sqrt{ax+b}$
$\frac{1}{e^x + e^{-x}}$	multiply by e^x , then $u = e^x$

Key formulas

$$\int \frac{du}{a^2 + u^2} = \frac{1}{a} \arctan \frac{u}{a}, \quad \int \frac{du}{\sqrt{a^2 - u^2}} = \arcsin \frac{u}{a}$$

Example Evaluate $\int_0^3 x\sqrt{x+1} dx$.

Solution $u = x + 1$, $x = u - 1$, limits 1 to 4: $\int_1^4 (u^{3/2} - u^{1/2}) du = [\frac{2}{5}u^{5/2} - \frac{2}{3}u^{3/2}]_1^4$.

$$\Rightarrow \frac{116}{15}.$$

Harder example Evaluate $\int_0^1 \frac{dx}{e^x + e^{-x}}$.

Solution Multiply top and bottom by e^x : $\frac{e^x}{1+e^{2x}}$, and $u = e^x$ gives $\int_1^e \frac{du}{1+u^2}$.

$$\Rightarrow \arctan e - \frac{\pi}{4}.$$

Traps du is not dx . Change the limits or convert back, never both. A constant factor ($du = 3x^2 dx$) must be divided out, not dropped.

Integration by parts

12

Method u is the factor that simplifies when differentiated (LIATE: log, inverse trig, algebraic, trig, exponential). Repeated parts on polynomial $\times$ exp/trig: use the table. If the integral comes back (cyclic), part twice and solve for I . A lone $\ln x$ or $\arctan x$: take $dv = dx$.

D		I
x^2	+	$\sin x$
$2x$	-	$-\cos x$
2	+	$-\sin x$
0	-	$\cos x$

Multiply along the arrows with alternating signs:
 $-x^2 \cos x + 2x \sin x + 2 \cos x$.

Key formulas

$$\int u dv = uv - \int v du$$

$$\int e^{ax} \sin bx dx = \frac{e^{ax}(a \sin bx - b \cos bx)}{a^2 + b^2}$$

Example Evaluate $\int_0^\pi x^2 \sin x dx$.

Solution From the table,

$$[-x^2 \cos x + 2x \sin x + 2 \cos x]_0^\pi = (\pi^2 - 2) - 2.$$

$$\Rightarrow \pi^2 - 4.$$

Harder example Evaluate $I = \int_0^{\pi/2} e^x \cos x dx$.

Solution Part twice with u the trigonometric factor both times: $I = [e^x \cos x] + [e^x \sin x] - I$, so $2I = [e^x(\cos x + \sin x)]_0^{\pi/2} = e^{\pi/2} - 1$.

$$\Rightarrow I = \frac{1}{2}(e^{\pi/2} - 1).$$

Example Evaluate $\int_1^e \ln^2 x dx$.

Solution $dv = dx$: $[x \ln^2 x] - 2 \int \ln x dx$, and again: antiderivative $x \ln^2 x - 2x \ln x + 2x$.

$$\Rightarrow e - 2.$$

Traps In a cyclic integral keep the *same* type of u on the second parting, or you undo the first. Signs alternate down the table.

Partial fractions and rational integrands

13

Method (1) If the top has degree $\geq$ the bottom, divide first. (2) Factor the bottom completely. (3) Write the template. (4) Cover-up for distinct linear factors; compare coefficients for the rest. (5) Integrate: logarithms, powers, and arctan after completing the square.

Key formulas

$$\frac{P}{(x-a)(x-b)} = \frac{A}{x-a} + \frac{B}{x-b}$$

$$\frac{P}{(x-a)^2} = \frac{A}{x-a} + \frac{B}{(x-a)^2}$$

$$\frac{P}{(x-a)(x^2+q)} = \frac{A}{x-a} + \frac{Bx+C}{x^2+q}$$

$$\int \frac{dx}{(x+h)^2+k^2} = \frac{1}{k} \arctan \frac{x+h}{k}$$

Problem type $\rightarrow$ first move

distinct linear factors	cover-up: set $x = a$
repeated factor	one term per power
irreducible quadratic	$Bx + C$ on top; split into $\ln$ and arctan
top degree too high	long division first
function of e^x	$u = e^x$, then partial fractions

Example Evaluate $\int_2^3 \frac{5x+1}{(x-1)(x+2)} dx$.

Solution Cover-up: $x = 1$ gives $A = \frac{6}{3} = 2$; $x = -2$ gives $B = \frac{-9}{-3} = 3$. So $[2 \ln |x-1| + 3 \ln |x+2|]_2^3 = 2 \ln 2 + 3 \ln 5 - 3 \ln 4$.

$$\Rightarrow \ln \frac{125}{16}.$$

Harder example Evaluate $\int_0^1 \frac{3x+1}{(x+1)(x^2+1)} dx$.

Solution $3x+1 = A(x^2+1) + (Bx+C)(x+1)$. $x = -1$: $A = -1$; x^2 : $A+B=0$, $B=1$; constant: $A+C=1$, $C=2$. So $-\frac{1}{x+1} + \frac{x}{x^2+1} + \frac{2}{x^2+1}$, giving $-\ln 2 + \frac{1}{2} \ln 2 + 2 \cdot \frac{\pi}{4}$.

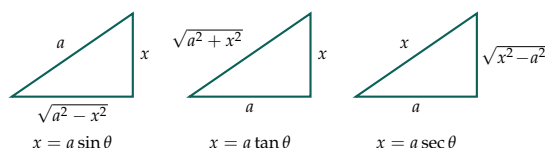
$$\Rightarrow \frac{\pi}{2} - \frac{1}{2} \ln 2.$$

Traps Cover-up only works for distinct linear factors. A repeated factor needs a term for every power.

Trigonometric integrals and substitutions

14

Method Odd power of $\sin$ or $\cos$: peel one off as the du , convert the even rest with $\sin^2 + \cos^2 = 1$. All even: power reduction. $\tan^m \sec^n$: even n , peel $\sec^2$; odd m , peel $\sec \tan$. Radicals: substitute from the triangle, then convert back with the same triangle.



Key formulas

$$\sin^2 x = \frac{1}{2}(1 - \cos 2x), \quad \cos^2 x = \frac{1}{2}(1 + \cos 2x)$$

$$\sin A \cos B = \frac{1}{2}[\sin(A+B) + \sin(A-B)]$$

Example Evaluate $\int_0^{\pi/2} \cos^5 x \, dx$.

Solution Peel one cosine: $\int (1 - \sin^2 x)^2 \cos x \, dx$, and $u = \sin x$ gives $\int_0^1 (1 - 2u^2 + u^4) \, du = 1 - \frac{2}{3} + \frac{1}{5} = \frac{8}{15}$.

Harder example Evaluate $\int_0^3 \frac{dx}{(9 + x^2)^{3/2}}$.

The Fundamental Theorem and accumulation

15

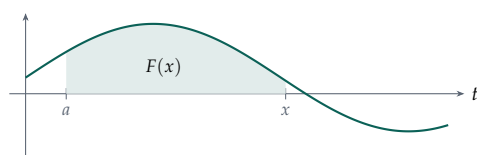
Method $F(x) = \int_a^x f(t) \, dt$ has $F' = f$ and $F(a) = 0$. So F rises where $f > 0$, has extremes where f changes sign, and is concave up where f is increasing. Values of F are signed areas under the graph of f . With variable limits, use the chain rule on each limit.

Key formulas

$$\frac{d}{dx} \int_{u(x)}^{v(x)} h(t) \, dt = h(v) v' - h(u) u'$$

Problem type → first move

variable limits	$h(v)v' - h(u)u'$
F from a graph of f	add signed areas
extremes of F	sign changes of f
$\frac{f_0^x}{x^n} \dots$ as $x \rightarrow 0$	L'Hopital, then FTC
integral equation	differentiate both sides



$F(x)$ is the signed area; area below the axis counts negative.

Example Differentiate $\int_x^{x^2} \ln t \, dt$ for $x > 0$.

Solution $\ln(x^2) \cdot 2x - \ln x \cdot 1 = 4x \ln x - \ln x = (4x - 1) \ln x$.

Harder example Find $\lim_{x \rightarrow 0} \frac{1}{x^3} \int_0^x \sin(t^2) \, dt$.

Solution It is $\frac{0}{0}$; L'Hopital and the FTC give $\frac{\sin(x^2)}{3x^2} \rightarrow \frac{1}{3}$.

Traps Forgetting the factor $v'(x)$. Area below the axis is negative in F .

Areas between curves

16

Method Sketch; find the intersections; integrate (top – bottom) dx , or (right – left) dy when the region is simpler sideways. Split the interval wherever the curves cross. Use symmetry to halve the work.

Key formulas

$$A = \int_a^b (f(x) - g(x)) \, dx = \int_c^d (x_R(y) - x_L(y)) \, dy$$

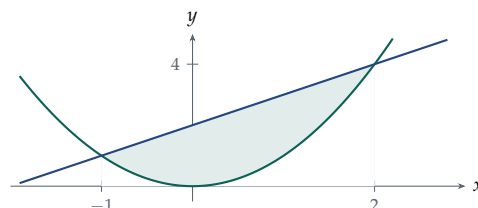
Problem type → first move

curves cross inside	split: $\int f - g $
bounded by $y^2 = \dots$	integrate in y
curve and its tangent	tangent below or above: one integral
line bisecting an area	area as a function of the parameter
parabola and a chord	area = $\frac{1}{6}(\text{width})^3$ for $y = x^2$

Solution $x = 3 \tan \theta$, $dx = 3 \sec^2 \theta \, d\theta$, $(9 + x^2)^{3/2} = 27 \sec^3 \theta$; θ runs from 0 to $\frac{\pi}{4}$: $\frac{1}{9} \int_0^{\pi/4} \cos \theta \, d\theta = \frac{1}{9} \sin \frac{\pi}{4} = \frac{\sqrt{2}}{18}$.

Traps dx changes too ($3 \sec^2 \theta \, d\theta$). $\sqrt{\cos^2 \theta} = |\cos \theta|$: check the range of θ .

Example Find the area between $y = x^2$ and $y = x + 2$.



Solution Intersections: $x^2 = x + 2$ at $x = -1, 2$. $\int_{-1}^2 (x + 2 - x^2) \, dx = \left[\frac{x^2}{2} + 2x - \frac{x^3}{3} \right]_{-1}^2 = \frac{10}{3} + \frac{7}{6} = \frac{9}{2}$.

Harder example Find the area bounded by $y^2 = 2x$ and $y = x - 4$.

Solution Sideways: x runs from $\frac{y^2}{2}$ to $y + 4$, and they meet at $y = -2, 4$. $\int_{-2}^4 (y + 4 - \frac{y^2}{2}) \, dy = \frac{40}{3} + \frac{14}{3} = 18$ (in x it would need two integrals).

Traps If the curves cross, $\int (f - g)$ subtracts areas. Check which curve is on top on each piece.

Volumes of revolution

17

Method Slice perpendicular to the axis: discs or washers, $\pi \int (R^2 - r^2)$. Slice parallel to the axis: shells, $2\pi \int (\text{radius})(\text{height})$. Radii are measured from the axis of rotation, not from zero. Known cross-sections: $V = \int A(x) \, dx$.

Key formulas

$$V_{\text{washer}} = \pi \int_a^b (R^2 - r^2) \, dx, \quad V_{\text{shell}} = 2\pi \int_a^b \rho h \, dx$$

Pappus: $V = 2\pi \bar{y} A$

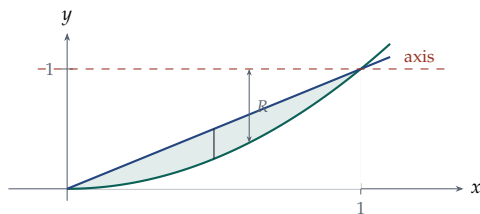
Problem type → first move

about an axis, region touches it	discs
gap between region and axis	washers, or shells
about $x = k$ or $y = k$	radius $ x - k $ or $ y - k $
region hard to slice one way	switch to shells
square or triangle sections	$\int A(x) \, dx$

Example Rotate the region under $y = x^2$, $0 \leq x \leq 1$, about the y -axis.

Solution Shells: $2\pi \int_0^1 x \cdot x^2 \, dx = \frac{\pi}{2}$. Check by washers in y : $\pi \int_0^1 (1 - y) \, dy = \frac{\pi}{2}$. $\Rightarrow \frac{\pi}{2}$ both ways.

Harder example Rotate the region between $y = x$ and $y = x^2$ about the line $y = 1$.



Solution Washers about $y = 1$: outer radius $R = 1 - x^2$, inner $r = 1 - x$.

$$\pi \int_0^1 [(1 - x^2)^2 - (1 - x)^2] dx = \pi \int_0^1 (x^4 - 3x^2 + 2x) dx \Rightarrow \frac{\pi}{5}.$$

Traps A washer is $\pi(R^2 - r^2)$, not $\pi(R - r)^2$. Rotating about $y = 1$: the radius is $1 - y$.

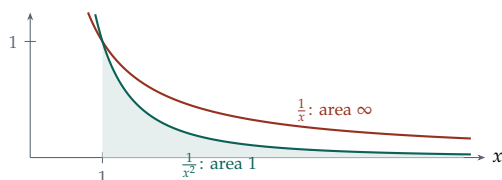
Arc length, surface area, improper integrals 18

Method Arc length: the problems are designed so that $1 + y'^2$ is a perfect square: expand it and look. *Improper*: replace the bad limit by a variable and take a limit; split at every interior singularity. *Convergence without evaluating*: compare with x^{-p} .

Key formulas

$$L = \int_a^b \sqrt{1 + y'^2} dx, \quad S = 2\pi \int_a^b y \sqrt{1 + y'^2} dx$$

$$\int_1^\infty \frac{dx}{x^p} < \infty \iff p > 1, \quad \int_0^1 \frac{dx}{x^p} < \infty \iff p < 1$$



Example Find the length of $y = \frac{x^3}{6} + \frac{1}{2x}$ from $x = 1$ to 2 .

Solution $y' = \frac{x^2}{2} - \frac{1}{2x^2}$, so $1 + y'^2 = (\frac{x^2}{2} + \frac{1}{2x^2})^2$ and $L = [\frac{x^3}{6} - \frac{1}{2x}]_1^2 \Rightarrow \frac{17}{12}$.

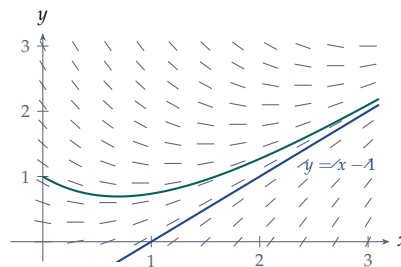
Harder example Does $\int_1^\infty \frac{\sin^2 x}{x^2} dx$ converge?

Solution $0 \leq \frac{\sin^2 x}{x^2} \leq \frac{1}{x^2}$ and $\int_1^\infty x^{-2} dx = 1$. $\Rightarrow$ yes, by comparison (its value is less than 1).

Traps $\int_{-1}^1 \frac{dx}{x^2}$ is not -2 : the integrand blows up at 0 and the integral diverges. Always look inside the interval.

Differential equations, slope fields, Euler 19

Method Classify first: *separable* ($y' = f(x)g(y)$), *linear* ($y' + Py = Q$, integrating factor), *logistic*. Fix the constant from the initial condition *before* rearranging, and state where the solution is valid. Slope field: segment of slope $f(x, y)$ at each grid point; isoclines $f = c$. Euler: $y_{n+1} = y_n + hf(x_n, y_n)$; concave up means Euler underestimates.



Slope field of $y' = x - y$. Every solution approaches $y = x - 1$.

Example Solve $y' = xy^2$, $y(0) = 1$.

Solution $-\frac{1}{y} = \frac{x^2}{2} + C$, and $y(0) = 1$ gives $C = -1$: $y = \frac{2}{2 - x^2}$.

$\Rightarrow y = \frac{2}{2 - x^2}$, valid for $|x| < \sqrt{2}$.

Harder example Solve $y' - 2y = e^{3x}$, $y(0) = 2$.

Solution $\mu = e^{-2x}$: $(ye^{-2x})' = e^x$, so $ye^{-2x} = e^x + C$ and $C = 1$.

$\Rightarrow y = e^{3x} + e^{2x}$.

Example Euler, $h = 0.5$, for $y' = 2x - y$, $y(0) = 1$: estimate $y(1)$.

Solution $y_1 = 1 + 0.5(0 - 1) = 0.5$; $y_2 = 0.5 + 0.5(1 - 0.5) = 0.75$. Exact $y = 2x - 2 + 3e^{-x}$ gives $y(1) = 1.104$; $y'' = 2 - y' > 0$ here, so Euler undershoots.

$\Rightarrow 0.75$, an underestimate.

Traps Dividing by y loses the solution $y = 0$. The constant goes in before you exponentiate or invert.

Sequences and series 20

Method First the n th-term test: if $a_n \not\rightarrow 0$, it diverges. Exact sums: geometric, telescoping (partial fractions), or differentiate a geometric series. Convergence: pick the test from the shape of the term (table). Recursive sequences: bounded + monotone $\Rightarrow$ converges, then solve $L = g(L)$.

Problem type $\rightarrow$ first move

r^n	geometric: $ r < 1$
rational in n	compare with n^{-p} , $p = \deg$ bottom – deg top
$n!$, c^n , n^n	ratio test
$(-1)^n b_n$	alternating: $b_n \downarrow 0$
$\frac{1}{n \ln n}$	integral test
$\frac{1}{n(n+k)}$	telescope

Key formulas

$$\sum_{n=0}^\infty ar^n = \frac{a}{1-r}, \quad \sum_{n=1}^\infty nx^n = \frac{x}{(1-x)^2} \quad (|x| < 1)$$

alternating: $|S - S_N| \leq b_{N+1}$

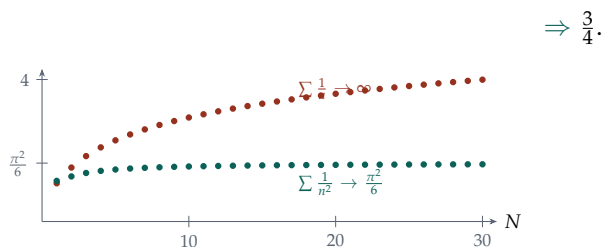
Example Find $\sum_{n=1}^\infty \frac{n}{3^n}$.

Solution $x = \frac{1}{3}$ in $\frac{x}{(1-x)^2}$: $\frac{1/3}{4/9}$.

$\Rightarrow \frac{3}{4}$.

Harder example Find $\sum_{n=2}^\infty \frac{1}{n^2 - 1}$.

Solution $\frac{1}{n^2 - 1} = \frac{1}{2}(\frac{1}{n-1} - \frac{1}{n+1})$; in the partial sums everything cancels except $\frac{1}{2}(1 + \frac{1}{2})$ and two terms that tend to 0.



Terms tending to 0 is not enough: the harmonic sums grow like $\ln N$.

Traps $a_n \rightarrow 0$ does not imply convergence. The ratio test says nothing when the limit is 1.

Power series and Taylor approximation

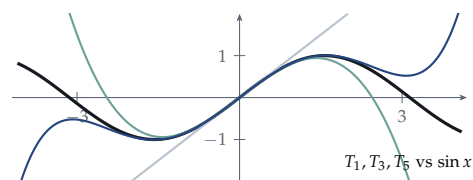
21

Method Build new series from known ones: substitute ($x \mapsto -x^2$), multiply, differentiate, integrate term by term. Radius from the ratio test; *test both endpoints separately*. Error: alternating-series bound, or Lagrange $\frac{M|x-a|^{n+1}}{(n+1)!}$.

Key formulas

$$\begin{aligned} e^x &= \sum \frac{x^n}{n!} & \frac{1}{1-x} &= \sum x^n \quad (|x| < 1) \\ \sin x &= \sum \frac{(-1)^n x^{2n+1}}{(2n+1)!} & \cos x &= \sum \frac{(-1)^n x^{2n}}{(2n)!} \\ \ln(1+x) &= \sum \frac{(-1)^{n+1} x^n}{n} & \arctan x &= \sum \frac{(-1)^n x^{2n+1}}{2n+1} \end{aligned}$$

$$(1+x)^k = 1 + kx + \frac{k(k-1)}{2!}x^2 + \dots \quad (|x| < 1)$$



Example Use $\ln \frac{1+x}{1-x} = 2(x + \frac{x^3}{3} + \frac{x^5}{5} + \dots)$ to compute $\ln 2$.

Solution $\frac{1+x}{1-x} = 2$ at $x = \frac{1}{3}$: $2(\frac{1}{3} + \frac{1}{81} + \frac{1}{1215}) = 0.69300\dots$; the true value is 0.69315.

$\Rightarrow$ three terms, error below 2×10^{-4} .

Harder example Find the interval of convergence of

$$\sum_{n \geq 1} \frac{(x-2)^n}{n 3^n}.$$

Solution Ratio: $\frac{|x-2|}{3} < 1$, so $-1 < x < 5$. At $x = 5$: $\sum \frac{1}{n}$ diverges; at $x = -1$: $\sum \frac{(-1)^n}{n}$ converges.

$\Rightarrow [-1, 5)$.

Traps The radius does not decide the endpoints. Centre at a : powers of $(x-a)$, not of x .

Part III · Algebra, trigonometry, geometry and discrete

BLOCKS 22–36

Polar coordinates: curves, slopes and areas

22

Method Convert with $x = r \cos \theta$, $y = r \sin \theta$, $r^2 = x^2 + y^2$. Sketch by tabulating r at multiples of $\frac{\pi}{6}$ and noting where $r = 0$ (tangent lines at the pole). For areas, sweep θ so each piece is traced *once*. Intersections: solve $r_1 = r_2$, then check the pole and $(-r, \theta + \pi)$ separately.

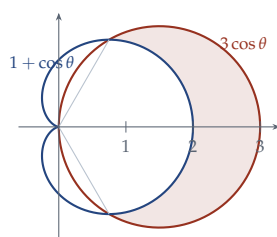
Key formulas

$$\frac{dy}{dx} = \frac{r' \sin \theta + r \cos \theta}{r' \cos \theta - r \sin \theta}, \quad A = \frac{1}{2} \int_{\alpha}^{\beta} r^2 d\theta$$

$$L = \int_{\alpha}^{\beta} \sqrt{r^2 + (r')^2} d\theta$$

Problem type → first move

convert to Cartesian	multiply through by r
area of one petal of $\cos n\theta$	between consecutive zeros of r
area between two curves	$\frac{1}{2} \int (r_{\text{out}}^2 - r_{\text{in}}^2)$
horizontal tangent	numerator of $\frac{dy}{dx}$ zero
points of intersection	solve, then test the pole



Inside $r = 3 \cos \theta$, outside the cardioid: θ runs over $[-\frac{\pi}{3}, \frac{\pi}{3}]$.

Example Find the area enclosed by the cardioid $r = 1 + \cos \theta$.

Solution $\frac{1}{2} \int_0^{2\pi} (1 + 2 \cos \theta + \cos^2 \theta) d\theta = \frac{1}{2} (2\pi + 0 + \pi) = \frac{3\pi}{2}$.

Harder example Find the area inside $r = 3 \cos \theta$ and outside $r = 1 + \cos \theta$.

Solution Intersect: $\cos \theta = \frac{1}{2}$, $\theta = \pm \frac{\pi}{3}$. By symmetry

$$A = \int_0^{\pi/3} (8 \cos^2 \theta - 2 \cos \theta - 1) d\theta = \left[3\theta + 2 \sin 2\theta - 2 \sin \theta \right]_0^{\pi/3}.$$

$\Rightarrow \pi$.

Traps $r = 3 \cos \theta$ is traced once on $[0, \pi]$, so $\int_0^{2\pi}$ doubles its area. Curves can meet at the pole at *different* values of θ .

Polynomials: remainder, factor, roots and coefficients

23

Method Remainder on dividing by $x - a$ is $p(a)$; by a quadratic it is $ax + b$, found by substituting the quadratic's roots (use p' for a repeated root). For symmetric functions of the roots, never find the roots: express everything in e_1, e_2, e_3 from the coefficients. Real coefficients: non-real roots come in conjugate pairs.

Key formulas

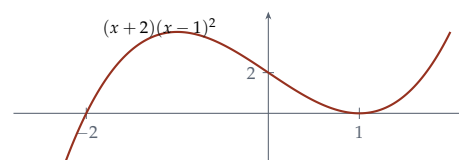
$$ax^3 + bx^2 + cx + d: \quad \sum \alpha = -\frac{b}{a}, \quad \sum \alpha\beta = \frac{c}{a}, \quad \alpha\beta\gamma = -\frac{d}{a}$$

$$\sum \alpha^2 = e_1^2 - 2e_2, \quad \sum \frac{1}{\alpha} = \frac{e_2}{e_3}$$

$$\sum \alpha^3 = e_1^3 - 3e_1e_2 + 3e_3$$

Problem type → first move

two remainders given	two equations in the unknowns
remainder on a quadratic	substitute its roots
repeated root	$p(a) = p'(a) = 0$
new roots $2\alpha, \alpha + 1$	substitute $x \mapsto \frac{x}{2}, x - 1$
rational roots	$\pm \frac{\text{factor of } d}{\text{factor of } a}$



A single root crosses the axis; a double root touches it.

Example α, β, γ are the roots of $x^3 - 2x^2 + 3x - 5 = 0$. Find $\sum \alpha^2$ and $\sum \frac{1}{\alpha}$.

Solution $e_1 = 2, e_2 = 3, e_3 = 5$. $\sum \alpha^2 = 4 - 6; \sum \frac{1}{\alpha} = \frac{3}{5}$.

$\Rightarrow -2$ and $\frac{3}{5}$ (a negative sum of squares: two roots are non-real).

Harder example Find the remainder when x^{100} is divided by $(x - 1)^2$.

Solution $x^{100} = Q(x)(x - 1)^2 + ax + b$. At $x = 1$: $a + b = 1$. Differentiate, then $x = 1$: $100 = a$.

$\Rightarrow 100x - 99$.

Traps The remainder on a quadratic is linear, not a constant. $\alpha\beta\gamma = -\frac{d}{a}$ carries a minus sign for a cubic.

Inequalities, surds and algebraic manipulation

24

Method Never multiply an inequality by an expression of unknown sign. Move everything to one side, combine into a single fraction, factor, and read the answer off a *sign diagram*. For absolute values, split at the zeros of each $|\cdot|$. To prove an inequality, try AM–GM, a square ≥ 0 , or Cauchy–Schwarz.

Key formulas

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b \geq 0), \quad (\sum a_i b_i)^2 \leq \sum a_i^2 \sum b_i^2$$

$$|x - a| < r \iff a - r < x < a + r$$

$$\sqrt{a + 2\sqrt{b}} = \sqrt{m} + \sqrt{n} \quad \text{if } m + n = a, mn = b$$

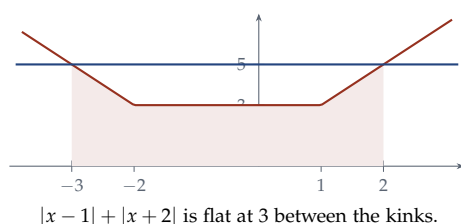
Problem type → first move

rational inequality	one fraction, sign diagram
sums of $ \cdot $	split into intervals
minimise $x + \frac{k}{x}$	AM-GM, equality at $x = \sqrt{k}$
nested surd	$\sqrt{m} + \sqrt{n}$ and match
rationalise	multiply by the conjugate

Example Solve $\frac{x+1}{x-2} \geq 3$.

Solution $\frac{x+1-3(x-2)}{x-2} = \frac{7-2x}{x-2} \geq 0$. Sign diagram with zeros at 2 (excluded) and $\frac{7}{2}$.

$$\Rightarrow 2 < x \leq \frac{7}{2}.$$



Harder example Solve $|x-1| + |x+2| < 5$.

Solution $x \geq 1$: $2x+1 < 5$, so $x < 2$. $-2 \leq x < 1$: $3 < 5$, always. $x < -2$: $-2x-1 < 5$, so $x > -3$.

$$\Rightarrow -3 < x < 2.$$

Surd check: $\sqrt{7+4\sqrt{3}} = \sqrt{7+2\sqrt{12}} = 2 + \sqrt{3}$ since $4+3=7$ and $4 \cdot 3 = 12$.

Traps Cross-multiplying $\frac{x+1}{x-2} \geq 3$ loses the case $x < 2$. Squaring both sides is safe only when both are non-negative.

Functions: composition, inverses and transformations 25

Method $f \circ g$ needs the range of g inside the domain of f . An inverse exists only on a one-to-one piece: restrict, swap x and y , solve; the domain of f^{-1} is the range of f . Transformations: changes *inside* act on x and go the opposite way; changes *outside* act on y as written.

Key formulas

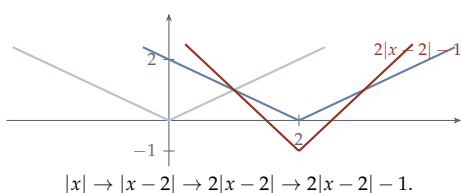
$$y = af(b(x-h)) + k:$$

stretch $\frac{1}{b}$ in x , a in y , then shift by (h, k)

$$f(f^{-1}(x)) = x, \quad (f \circ g)^{-1} = g^{-1} \circ f^{-1}$$

Problem type → first move

domain of $f \circ g$	x with $g(x) \in \text{dom } f$
$f = f^{-1}$ intersections	for increasing f , solve $f(x) = x$
$y = f(x) $	reflect the negative parts up
$y = f(x)$	keep $x \geq 0$, mirror it
self-inverse	graph symmetric in $y = x$



Example Find the inverse of $f(x) = \frac{2x+1}{x-3}$.

Solution $y(x-3) = 2x+1 \Rightarrow x(y-2) = 3y+1$.
 $\Rightarrow f^{-1}(x) = \frac{3x+1}{x-2}, x \neq 2$.

Harder example $f(x) = x^2 - 4x + 1$ for $x \geq 2$. Find f^{-1} and where the graphs of f and f^{-1} meet.

Solution $f(x) = (x-2)^2 - 3$, range $[-3, \infty)$, so $f^{-1}(x) = 2 + \sqrt{x+3}$ for $x \geq -3$. f is increasing, so the graphs meet on $y = x$: $x^2 - 5x + 1 = 0$ with $x \geq 2$.

$$\Rightarrow x = y = \frac{5+\sqrt{21}}{2}.$$

Traps $y = f(2x)$ is a compression by $\frac{1}{2}$. A decreasing f can meet f^{-1} off the line $y = x$.

Exponentials and logarithms

26

Method Reduce to one base. Equations in a^{2x} and a^x are quadratics in $u = a^x$ (reject $u \leq 0$). Mixed bases: take $\ln$ of both sides. Mixed log bases: change to one base. Always check that every log argument is positive in the answer.

Key formulas

$$\log_a xy = \log_a x + \log_a y, \quad \log_a x^k = k \log_a x$$

$$\log_a x = \frac{\ln x}{\ln a}, \quad \log_{a^n} x = \frac{1}{n} \log_a x, \quad a^{\log_a x} = x$$

Problem type → first move

$a^{2x} + pa^x + q = 0$	quadratic in $u = a^x > 0$
$a^x = b^{x+c}$	take logs, collect x
$\log_a x + \log_{a^2} x$	change to base a
growth or decay	$N = N_0 e^{kt}$, k from two data
straight-line graph	$\ln y$ against x or $\ln x$

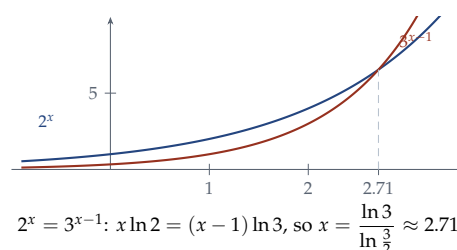
Example Solve $4^x - 3 \cdot 2^{x+1} + 8 = 0$.

Solution $u = 2^x$: $u^2 - 6u + 8 = 0$, so $u = 2$ or 4 .

$$\Rightarrow x = 1 \text{ or } x = 2.$$

Harder example Solve $\log_2 x + \log_4 x + \log_8 x = 11$.

Solution In base 2: $\log_2 x (1 + \frac{1}{2} + \frac{1}{3}) = 11$, so $\log_2 x = 6$.
 $\Rightarrow x = 64$.



Traps $\ln(a+b) \neq \ln a + \ln b$. Dividing by $\ln x$ loses $x = 1$. A root $u < 0$ of the quadratic gives no x .

Trigonometric identities and equations

27

Method Identities: work from the more complicated side, convert to $\sin$ and $\cos$, and aim for one function of one angle. Equations: reduce to a single function (use $\sin^2 + \cos^2 = 1$), factor, never divide by a factor that might be zero, then write the *general solution* and pick the values in the interval.

Key formulas

$$\sin x = \sin a \iff x = a + 2k\pi \text{ or } \pi - a + 2k\pi$$

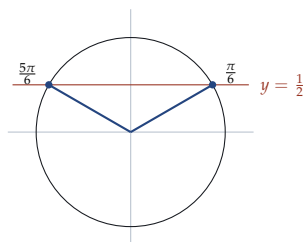
$$\cos x = \cos a \iff x = \pm a + 2k\pi$$

$$\tan x = \tan a \iff x = a + k\pi$$

$$1 + \tan^2 x = \sec^2 x, \quad 1 + \cot^2 x = \csc^2 x$$

Problem type → first move

quadratic in $\sin$ or $\cos$ swap $\cos^2 = 1 - \sin^2$, factor
 $\sin mx = \cos nx$ $\cos nx = \sin(\frac{\pi}{2} - nx)$
 sin and cos linear R-form (block 28)
 $\sin x = k \cos x$ divide by $\cos x$ (check $\cos x = 0$)
 interval $[0, 2\pi)$ for $3x$ solve for $3x$ in $[0, 6\pi)$


Compound and multiple angles and the R-form

28

Method Any $a \sin x + b \cos x$ is a single wave $R \sin(x + \alpha)$ with $R = \sqrt{a^2 + b^2}$. That turns “solve”, “maximum” and “range” questions into one-line reads. Multiple angles: expand $\sin 3x$, $\cos 3x$ into powers; to integrate powers, go the other way (power reduction).

Key formulas

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$$\sin 3x = 3 \sin x - 4 \sin^3 x, \quad \cos 3x = 4 \cos^3 x - 3 \cos x$$

$$a \sin x + b \cos x = R \sin(x + \alpha), \quad \tan \alpha = \frac{b}{a}$$

Problem type → first move

max or min of $\pm R$

$a \sin x + b \cos$

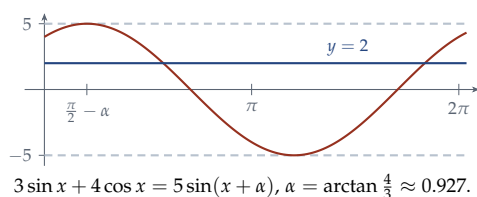
range of $\frac{1}{c + a \sin x + b \cos x}$ denominator in $[c - R, c + R]$

solve $\sin(x + \alpha) = \frac{k}{R}$

$a \sin x + b \cos x = k$

exact $\sin 15^\circ$, $\tan 75^\circ$ $45^\circ \pm 30^\circ$

$t = \tan \frac{x}{2}$ $\sin x = \frac{2t}{1+t^2}$, $\cos x = \frac{1-t^2}{1+t^2}$



Example Solve $3 \sin x + 4 \cos x = 2$ for $0 \leq x < 2\pi$.

Solution $5 \sin(x + \alpha) = 2$ with $\alpha \approx 0.9273$. $x + \alpha = 0.4115 + 2\pi$ or $\pi - 0.4115$.

$$\Rightarrow x \approx 1.803 \text{ or } x \approx 5.767.$$

Harder example Show that $\sin 18^\circ = \frac{\sqrt{5}-1}{4}$.

Solution Let $\theta = 18^\circ$. Then $2\theta = 90^\circ - 3\theta$, so $\sin 2\theta = \cos 3\theta$: $2 \sin \theta \cos \theta = 4 \cos^3 \theta - 3 \cos \theta$. Divide by $\cos \theta \neq 0$: $2s = 1 - 4s^2$, so $4s^2 + 2s - 1 = 0$; take the positive root.

$$\Rightarrow \sin 18^\circ = \frac{-1+\sqrt{5}}{4}.$$

Traps Check the quadrant of α from the signs of $R \cos \alpha = a$ and $R \sin \alpha = b$. The answers must be shifted back into the interval.

$\sin x = \frac{1}{2}$ has two solutions per turn: a and $\pi - a$.

Example Solve $2 \cos^2 x + 3 \sin x - 3 = 0$ for $0 \leq x < 2\pi$.

Solution $2(1 - \sin^2 x) + 3 \sin x - 3 = 0 \Rightarrow (2 \sin x - 1)(\sin x - 1) = 0$.

$$\Rightarrow x = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}.$$

Harder example Find the general solution of $\sin 3x = \cos x$.

Solution $\sin 3x = \sin(\frac{\pi}{2} - x)$. Either $3x = \frac{\pi}{2} - x + 2k\pi$, or $3x = \pi - (\frac{\pi}{2} - x) + 2k\pi$.

$$\Rightarrow x = \frac{\pi}{8} + \frac{k\pi}{2} \text{ or } x = \frac{\pi}{4} + k\pi.$$

Traps Cancelling $\sin x$ from $\sin x \cos x = \sin x$ loses $\sin x = 0$. Squaring introduces extra roots: check each one.

Solution of triangles and three-dimensional trigonometry

29

Method Sides and opposite angles: sine rule. Two sides and the included angle, or three sides: cosine rule. Given a, b and a non-included angle A , test the *ambiguous case*. In 3D, find a right-angled triangle containing the angle you want: the angle between a line and a plane uses the line's projection onto the plane.

Key formulas

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$$

$$c^2 = a^2 + b^2 - 2ab \cos C, \quad \text{Area} = \frac{1}{2}ab \sin C$$

$$\text{Heron: } \sqrt{s(s-a)(s-b)(s-c)}, \quad s = \frac{a+b+c}{2}$$

Problem type → first move

SAS or SSS

cosine rule first

SSA with $a < b$

ambiguous: B or $180^\circ - B$

bearings

draw north lines; alternate angles

angle line-plane

drop a perpendicular, use the

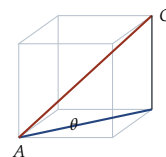
projection

angle between two planes

perpendiculars to the common line

Example In triangle ABC , $a = 7$, $b = 8$, $C = 60^\circ$. Find c and the area.

Solution $c^2 = 49 + 64 - 2 \cdot 56 \cdot \frac{1}{2} = 57$; area $= \frac{1}{2} \cdot 7 \cdot 8 \cdot \frac{\sqrt{3}}{2}$.
 $\Rightarrow c = \sqrt{57} \approx 7.55$, area $14\sqrt{3}$.



Space diagonal AG against its projection on the base.

Harder example A cube has edge 2. Find the angle between a space diagonal and the base.

Solution The projection of AG on the base is the face diagonal, length $2\sqrt{2}$; the rise is 2. $\tan \theta = \frac{2}{2\sqrt{2}}$.

$$\Rightarrow \theta = \arctan \frac{1}{\sqrt{2}} \approx 35.3^\circ.$$

Ambiguous case: $A = 30^\circ$, $a = 5$, $b = 8$ gives $\sin B = 0.8$, so $B \approx 53.1^\circ$ or 126.9° : two triangles.

Traps Calculators return only the acute B from $\sin^{-1}$. Using the sine rule for an obtuse angle first loses it: find the largest angle by the cosine rule.

Mathematical induction

30

Method Write $P(n)$ in full. *Base case:* check the first n . *Step:* assume $P(k)$, and write the $P(k+1)$ target down before you start, so you know what you are aiming at. Find $P(k)$ inside $P(k+1)$: split off the last term (sums), factor out the old expression (divisibility), or chain inequalities. Conclude in words.

Problem type → first move

sum formula	$S_{k+1} = S_k + u_{k+1}$, then factor
divisibility	$f(k+1) = c f(k) + (\text{multiple})$
inequality	$\text{LHS}_{k+1} > \dots \geq \text{RHS}_{k+1}$
recurrence	use u_k in g
$u_{n+1} = g(u_n)$	
derivatives $\frac{d^n}{dx^n}$	differentiate the $P(k)$ formula
matrix powers A^n	$A^{k+1} = A^k A$

Example Prove $3 \mid 4^n - 1$ for all $n \geq 1$.

Solution $n = 1$: $3 \mid 4^1 - 1 = 3$. If $4^k - 1 = 3m$, then $4^{k+1} - 1 = 4(4^k - 1) + 3 = 3(4m + 1)$.

⇒ true for all $n \geq 1$ by induction.

Harder example Prove $\sum_{r=1}^n r \cdot r! = (n+1)! - 1$.

Solution $n = 1$: $1 = 2! - 1$. Step: $(k+1)! - 1 + (k+1)(k+1)! = (k+1)!(k+2) - 1 = (k+2)! - 1$.

⇒ the formula for $n = k+1$.

Inequalities need a starting value: $2^n > n^2$ holds for $n \geq 5$, and the step uses $2k^2 \geq (k+1)^2$, which needs $k \geq 3$.

Traps Assuming $P(k+1)$ and working back to a truth proves nothing. Base case at the wrong n . “True for k ” must be the hypothesis, not what you verify.

Binomial theorem and expansions

31

Method Write the *general term* first, then collect the powers of x and solve for r . That handles specific coefficients, constant terms and “no such term” at once. For a product of two expansions, list the terms of each factor up to the power you need and pair them.

Key formulas

$$(a+b)^n = \sum_{r=0}^n \binom{n}{r} a^{n-r} b^r, \quad T_{r+1} = \binom{n}{r} a^{n-r} b^r$$

$$\sum \binom{n}{r} = 2^n, \quad \sum \binom{n}{r}^2 = \binom{2n}{n}, \quad \binom{n}{r} + \binom{n}{r+1} = \binom{n+1}{r+1}$$

Problem type → first move

coefficient of x^k	general term, solve for r
term independent of x	power of x equal to 0
product of two brackets	pair terms, add
approximate 1.02^{10}	$(1+0.02)^{10}$, first terms
greatest term	ratio T_{r+1}/T_r against 1

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Each entry is the sum of the two above it.

Example Find the coefficient of x^5 in $(2x - \frac{1}{x})^9$, and the term independent of x .

Solution $T_{r+1} = \binom{9}{r} (2x)^{9-r} (-x^{-1})^r = \binom{9}{r} 2^{9-r} (-1)^r x^{9-2r}$. x^5 : $r = 2$, $36 \cdot 2^7$. $9 - 2r = 0$ has no integer solution.

⇒ 4608; no constant term.

Harder example Find the coefficient of x^3 in $(1+x)^5(1-2x)^4$.

Solution $(1+x)^5$: 1, 5, 10, 10, 5, 1. $(1-2x)^4$: 1, -8, 24, -32. Pair to total power 3: $-32 + 5 \cdot 24 + 10 \cdot (-8) + 10$.

⇒ 18.

Traps T_{r+1} , not T_r , contains $\binom{n}{r}$. Keep the sign and power of each part: $(-\frac{1}{x})^r = (-1)^r x^{-r}$.

Complex numbers: Argand diagram, de Moivre and roots of unity

32

Method Use Cartesian form to add and to solve linear equations; polar form $re^{i\theta}$ to multiply, divide, raise to powers and take roots. The n th roots of w sit on a circle of radius $|w|^{1/n}$, spaced $\frac{2\pi}{n}$ apart. Loci: translate $|z-a|$ as “distance from a ”.

Key formulas

$$(re^{i\theta})^n = r^n e^{in\theta}, \quad z^n = Re^{i\phi} \iff z = R^{1/n} e^{i(\phi+2k\pi)/n}$$

$$z\bar{z} = |z|^2, \quad \cos \theta = \frac{1}{2}(z + z^{-1}) \text{ for } |z| = 1$$

$$1 + \omega + \dots + \omega^{n-1} = 0 \quad (\omega^n = 1, \omega \neq 1)$$

Problem type → first move

$(1+i)^n, (\sqrt{3}+i)^n$	polar form, de Moivre
$z^n = w$	one root, then rotate by $\frac{2\pi}{n}$
$\cos n\theta$ in powers	expand $(\cos \theta + i \sin \theta)^n$
$ z-a = z-b $	perpendicular bisector
$\arg(z-a) = \theta$	half-line from a

Example Compute $(1+i)^{10}$.

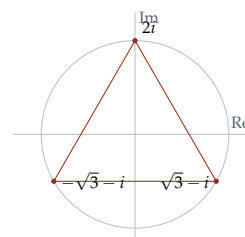
Solution $1+i = \sqrt{2}e^{i\pi/4}$, so $(1+i)^{10} = 2^5 e^{i10\pi/4} = 32e^{i5\pi/2}$.

⇒ 32i.

Harder example Solve $z^3 = -8i$.

Solution $-8i = 8e^{-i\pi/2}$, so $z = 2e^{i(-\pi/6+2k\pi/3)}$ with arguments $-\frac{\pi}{6}, \frac{\pi}{2}, \frac{7\pi}{6}$.

⇒ $\sqrt{3}-i, 2i, -\sqrt{3}-i$.



The three cube roots of $-8i$: an equilateral triangle on $|z| = 2$.

Traps $\arg(-1-i) = -\frac{3\pi}{4}$, not $\arctan 1$: check the quadrant. Forgetting the $+2k\pi$ gives only one root.

Matrices and determinants of order two and three

33

Method A 2×2 matrix is a map of the plane: its columns are the images of $\mathbf{i}$ and $\mathbf{j}$, and $|\det|$ is the area scale factor. For 3×3 determinants, expand along the row or column with the most zeros, after row operations to create zeros. A is invertible exactly when $\det A \neq 0$.

Key formulas

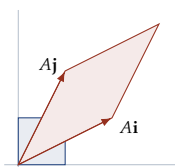
$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

$$\det(AB) = \det A \det B, \quad (AB)^{-1} = B^{-1}A^{-1}$$

$$A^{-1} = \frac{1}{\det A} \operatorname{adj} A, \quad \det(kA) = k^n \det A$$

Problem type → first move

singular for which k $\det = 0$, solve
 image of a shape multiply each vertex
 area of the image $|\det| \times \text{original area}$
 A^n pattern, then induction
 solve $AX = B$ $X = A^{-1}B$ (order matters)



$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ sends the unit square to a parallelogram of area $\det A = 3$.

Example Find the inverse of $\begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix}$.

Solution $\det = 6 - 5 = 1$.

$$\Rightarrow \begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix}.$$

Harder example For which k is $\begin{pmatrix} 1 & 2 & k \\ 0 & k & 1 \\ 1 & 1 & 1 \end{pmatrix}$ singular?

Solution Along row 1: $1(k - 1) - 2(0 - 1) + k(0 - k) = -k^2 + k + 1$. Set it to 0.

$$\Rightarrow k = \frac{1 \pm \sqrt{5}}{2}.$$

Traps $AB \neq BA$ in general, so $AX = B$ gives $X = A^{-1}B$, not BA^{-1} . The middle cofactor carries a minus sign.

Systems of linear equations

34

Method Row-reduce the augmented matrix to echelon form. A pivot in every variable column: one solution. A row $0 = c \neq 0$: no solution. Otherwise free variables: infinitely many, written with a parameter. With a parameter k in the coefficients, the special values are where the last pivot vanishes; then check the right-hand side.

Key formulas

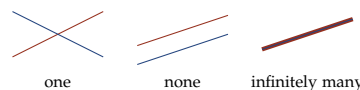
$\det A \neq 0 \iff$ unique solution

$$x_i = \frac{\det A_i}{\det A} \quad (\text{Cramer})$$

$\det A = 0$: none or infinitely many

Problem type → first move

three numeric equations eliminate one variable twice
 parameter in coefficients echelon form, last pivot = 0
 consistency condition last row $0 = \dots$ must vanish
 three planes unique point, line, sheaf, or prism
 homogeneous $Ax = 0$ non-trivial iff $\det A = 0$



In two variables: crossing, parallel, or the same line.

Example Solve $x + y + z = 6$, $2x - y + z = 3$, $x + 2y - z = 2$.

Solution Add the first and third: $2x + 3y = 8$. Add the second and third: $3x + y = 5$. So $x = 1$, $y = 2$, then $z = 3$.

$$\Rightarrow (1, 2, 3).$$

Harder example Discuss $x + y + z = 1$, $x + 2y + 3z = 2$, $x + 3y + kz = m$.

Solution $R_2 - R_1$: $y + 2z = 1$. $R_3 - R_1 - 2(R_2 - R_1)$: $(k - 5)z = m - 3$.

$\Rightarrow k \neq 5$: unique. $k = 5$, $m = 3$: a line of solutions. $k = 5$, $m \neq 3$: none.

Traps Dividing by $k - 5$ without saying $k \neq 5$. A zero determinant alone does not decide between “none” and “infinitely many”.

Vectors, lines and planes in space

35

Method Dot product for angles, perpendicularity and projections; cross product for normals and areas. A plane is a point plus a normal: $\mathbf{n} \cdot \mathbf{r} = \mathbf{n} \cdot \mathbf{a}$. For intersections, substitute the line's parametric form into the plane or the other line. Distances are projections onto a normal.

Key formulas

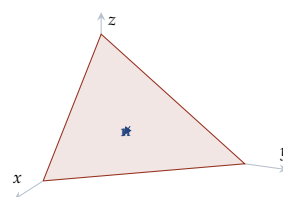
$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos \theta, \quad |\mathbf{a} \times \mathbf{b}| = |\mathbf{a}||\mathbf{b}| \sin \theta$$

$$d(P, \Pi) = \frac{|ax_0 + by_0 + cz_0 - d|}{\sqrt{a^2 + b^2 + c^2}}$$

$$d(\text{skew lines}) = \frac{|(\mathbf{b} - \mathbf{a}) \cdot (\mathbf{u} \times \mathbf{v})|}{|\mathbf{u} \times \mathbf{v}|}$$

Problem type → first move

plane through three points $\mathbf{n} = \overrightarrow{AB} \times \overrightarrow{AC}$
 line meets plane substitute $\mathbf{r}(t)$, solve for t
 angle line-plane $\sin \theta = \frac{|\mathbf{u} \cdot \mathbf{n}|}{|\mathbf{u}||\mathbf{n}|}$
 two lines parallel, meet, or skew
 foot of perpendicular line along $\mathbf{n}$ through P
 volume of tetrahedron $\frac{1}{6} |\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|$



The plane through the intercepts 1, 2, 3, with normal $(6, 3, 2)$.

Example Find the plane through $(1, 0, 0)$, $(0, 2, 0)$, $(0, 0, 3)$ and its distance from O .

Solution $\vec{AB} \times \vec{AC} = (-1, 2, 0) \times (-1, 0, 3) = (6, 3, 2)$.
 Plane $6x + 3y + 2z = 6$; distance $\frac{6}{\sqrt{49}}$.
 $\Rightarrow 6x + 3y + 2z = 6$; distance $\frac{6}{7}$.

Harder example Find the distance between $\mathbf{r} = (1, 0, 0) + s(1, 1, 0)$ and $\mathbf{r} = (0, 1, 3) + t(0, 1, 1)$.

Solution $\mathbf{u} \times \mathbf{v} = (1, -1, 1)$; $\mathbf{b} - \mathbf{a} = (-1, 1, 3)$; dot product 1.

$$\Rightarrow \frac{1}{\sqrt{3}}.$$

Traps The angle between a line and a plane uses sin, not cos. Lines that are not parallel may still not meet in 3D: check all three coordinates.

Counting, permutations and probability

36

Method Decide: does order matter, and are repeats allowed? Handle restrictions first (“must be together”: glue into one block; “not together”: total minus together, or gaps). “At least one”: $1 - P(\text{none})$. Conditional probability: draw a tree, or restrict the sample space. Reverse conditioning is Bayes.

Key formulas

$${}_nP_r = \frac{n!}{(n-r)!}, \quad \binom{n}{r} = \frac{n!}{r!(n-r)!}$$

$$\frac{n!}{p!q!\dots} \text{ (with repeats),} \quad P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$X \sim B(n, p): P(X = r) = \binom{n}{r} p^r (1-p)^{n-r}, \quad E = np$$

Problem type $\rightarrow$ first move

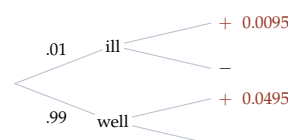
letters with repeats	$\frac{n!}{\text{repeats!}}$
circular arrangements	$(n-1)!$
committee with conditions	cases, or total minus bad
“not adjacent”	arrange the others, fill gaps
test positive: ill?	Bayes, tree diagram

Example How many arrangements of MISSISSIPPI are there?

Solution 11 letters: I $\times$ 4, S $\times$ 4, P $\times$ 2. $\frac{11!}{4!4!2!}$.

 $\Rightarrow 34650$.

Harder example A disease affects 1%. A test detects it 95% of the time and gives false positives 5% of the time. Find $P(\text{ill} | +)$.



Solution $P(+) = 0.01(0.95) + 0.99(0.05) = 0.059$, so $P(\text{ill} | +) = \frac{0.0095}{0.059}$.

 $\Rightarrow \approx 0.161$: most positives are false.

Traps Double-counting in “at least one” by choosing one first, then any. $P(A|B) \neq P(B|A)$.

Part IV · Multivariable calculus

BLOCKS 37–42

Partial derivatives, the chain rule and implicit differentiation

37

Method f_x : differentiate in x with every other variable held constant. It is the slope of the trace of the surface in the plane $y = \text{const}$. Chain rule: add one term for every path from the output to the variable (a tree diagram). Implicit: treat $F(x, y, z) = 0$ as defining z and use the formula, not repeated rearranging.

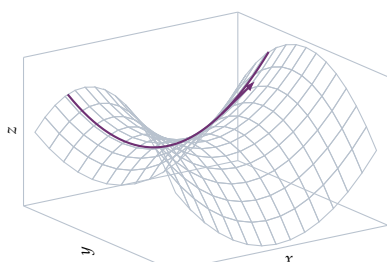
Key formulas

$$\begin{aligned} \frac{dz}{dt} &= f_x \frac{dx}{dt} + f_y \frac{dy}{dt}, & \frac{\partial z}{\partial s} &= f_x \frac{\partial x}{\partial s} + f_y \frac{\partial y}{\partial s} \\ \frac{dy}{dx} &= -\frac{F_x}{F_y}, & \frac{\partial z}{\partial x} &= -\frac{F_x}{F_z}, & \frac{\partial z}{\partial y} &= -\frac{F_y}{F_z} \end{aligned}$$

$$f_{xy} = f_{yx} \text{ (Clairaut, continuous second partials)}$$

Problem type → first move

all second partials $z = f(x(t), y(t))$ $z = f(x, y, z) = 0$ verify a PDE (Laplace, wave)
four, check $f_{xy} = f_{yx}$
one-variable chain rule, two terms
 $x = r \cos \theta$, tree diagram
 $-F_x/F_z$ compute, substitute, simplify



The saddle $z = x^2 - y^2$: f_x at a point is the slope of the trace $y = \text{const}$.

Example For $f = x^2y + e^{xy}$, find f_x and f_{xy} .

Solution $f_x = 2xy + ye^{xy}$. Differentiate in y : $2x + e^{xy} + xye^{xy}$.

$$\Rightarrow f_{xy} = 2x + (1 + xy)e^{xy}.$$

Harder example $w = x^2 + y^2 + z^2$ with $x = st$, $y = s \cos t$, $z = s \sin t$. Find $\partial w / \partial s$.

Solution $2x \cdot t + 2y \cos t + 2z \sin t = 2st^2 + 2s(\cos^2 t + \sin^2 t)$. Check: $w = s^2(t^2 + 1)$ directly.

$$\Rightarrow 2s(t^2 + 1).$$

Implicit: $x^3 + y^3 + z^3 + 6xyz = 1$ gives $\frac{\partial z}{\partial x} = -\frac{x^2 + 2yz}{z^2 + 2xy}$.

Traps The minus sign in $-F_x/F_z$. In the chain rule, evaluate every partial at the *same* point.

Gradients, directional derivatives and tangent planes

38

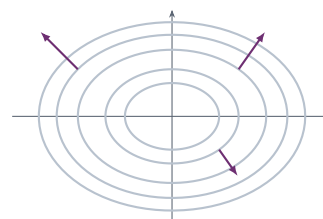
Method ∇f points in the direction of fastest increase, has length equal to that rate, and is *perpendicular to level sets*. The directional derivative is the component of ∇f along a *unit* vector. So a tangent plane to $F(x, y, z) = c$ has normal ∇F , and the tangent plane to $z = f(x, y)$ is the linearisation.

Key formulas

$$\begin{aligned} D_{\mathbf{u}}f &= \nabla f \cdot \mathbf{u}, \quad |\mathbf{u}| = 1, & \max D_{\mathbf{u}}f &= |\nabla f| \\ \nabla F(P) \cdot (\mathbf{r} - P) &= 0 \text{ (tangent plane to } F = c) \\ z &\approx f(a, b) + f_x(x - a) + f_y(y - b) \end{aligned}$$

Problem type → first move

rate in direction $\mathbf{v}$ $\mathbf{u} \perp \nabla f$ (along the contour)
steepest ascent ∇F is the normal
zero rate $\mathbf{u} \perp \nabla f$ (along the contour)
tangent plane, normal ∇F is the normal
line
estimate $f(3.02, 3.97)$ linearise at $(3, 4)$



Contours of $x^2 + 2y^2$: every gradient crosses its contour at right angles.

Example $f = x^2y + 3yz$. Find the rate of change at $(1, 2, -1)$ in the direction of $(2, -1, 2)$, and the maximum rate.

Solution $\nabla f = (2xy, x^2 + 3z, 3y) = (4, -2, 6)$. $\mathbf{u} = \frac{1}{3}(2, -1, 2)$: $\frac{1}{3}(8 + 2 + 12)$. Maximum $|\nabla f| = \sqrt{56}$.
 $\Rightarrow \frac{22}{3}$; maximum $2\sqrt{14}$.

Harder example Find the tangent plane to $x^2 + 2y^2 + 3z^2 = 6$ at $(1, 1, 1)$, and estimate $\sqrt{3.02^2 + 3.97^2}$.

Solution $\nabla F = (2, 4, 6)$: $2(x - 1) + 4(y - 1) + 6(z - 1) = 0$. For the estimate, $f = \sqrt{x^2 + y^2}$ at $(3, 4)$: $5 + \frac{3}{5}(0.02) + \frac{4}{5}(-0.03)$.

$$\Rightarrow x + 2y + 3z = 6; \approx 4.988.$$

Traps Using $\mathbf{v}$ instead of the unit vector. The gradient of $f(x, y)$ lives in the plane; the normal to the graph is $(f_x, f_y, -1)$.

Extrema, the second-derivative test and Lagrange multipliers

39

Method Free extrema: solve $f_x = f_y = 0$, then classify each point with D . On a closed bounded region, compare interior critical values, boundary extremes (parametrise each edge) and corners. With a constraint $g = c$, solve $\nabla f = \lambda \nabla g$ with $g = c$ and compare the values: the largest is the maximum.

Key formulas

$$D = f_{xx}f_{yy} - f_{xy}^2$$

$$D > 0, f_{xx} > 0: \text{min}; \quad D > 0, f_{xx} < 0: \text{max}$$

$$D < 0: \text{saddle}; \quad D = 0: \text{no conclusion}$$

$$\nabla f = \lambda \nabla g, \quad g = c \quad (\text{two constraints: } + \mu \nabla h)$$

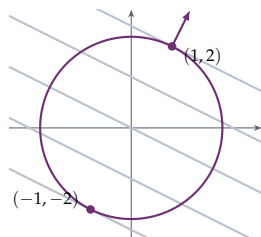
Problem type → first move

classify critical points	table of (x, y) , D , f_{xx}
closed region	interior, edges, corners
one constraint	$\nabla f = \lambda \nabla g$, eliminate λ
distance to a surface	minimise the squared distance
box of fixed volume	Lagrange, or substitute

Example Classify the critical points of $f = x^3 - 3x + y^2 - 4y$.

Solution $f_x = 3x^2 - 3 = 0$, $f_y = 2y - 4 = 0$: $(\pm 1, 2)$.
 $f_{xx} = 6x$, $f_{yy} = 2$, $f_{xy} = 0$, so $D = 12x$.

⇒ $(1, 2)$: minimum, $f = -6$. $(-1, 2)$: saddle.



Level lines $x + 2y = c$ against $x^2 + y^2 = 5$: extremes where they are tangent, $\nabla f \parallel \nabla g$.

Harder example Find the extreme values of $f = x + 2y$ on $x^2 + y^2 = 5$.

Solution $1 = 2\lambda x$, $2 = 2\lambda y$, so $y = 2x$. Then $5x^2 = 5$, $x = \pm 1$.

⇒ maximum 5 at $(1, 2)$; minimum -5 at $(-1, -2)$.

Traps $D = 0$ decides nothing. Lagrange finds candidates, not answers: evaluate f at each. Dividing by a variable that could be 0 loses solutions.

Vector fields, line integrals, conservative fields and Green's theorem

41

Double and triple integrals, coordinate changes and the Jacobian

40

Method Sketch the region first. Choose the order (or coordinates) that makes the limits simple, and reverse the order when the inner integral is impossible. Circles: polar. Cylinders: cylindrical. Spheres and cones: spherical. Other regions: pick u, v from the boundary equations and multiply by $|J|$.

Key formulas

$$dA = r dr d\theta, \quad dV = r dz dr d\theta$$

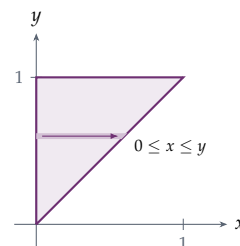
$$dV = \rho^2 \sin \phi d\rho d\phi d\theta$$

$$dx dy = \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du dv$$

$$\frac{\partial(x, y)}{\partial(u, v)} = 1 / \frac{\partial(u, v)}{\partial(x, y)}$$

Problem type → first move

$\int e^{y^2} dy$ inside	reverse the order
disc or annulus	polar, extra factor r
volume between surfaces	$\iint (\text{top} - \text{bottom}) dA$
sphere, cone, ice cream	spherical
region bounded by $x \pm y = c$	$u = x + y, v = x - y, J = \frac{1}{2}$
mass, centroid	$\iint \rho dA, \bar{x} = \frac{1}{M} \iint x\rho dA$



The same triangle as $x \leq y \leq 1$ (vertical strips) or $0 \leq x \leq y$ (horizontal strips).

Example Evaluate $\int_0^1 \int_x^1 e^{y^2} dy dx$.

Solution Reverse: $\int_0^1 \int_0^y e^{y^2} dx dy = \int_0^1 y e^{y^2} dy$.

⇒ $\frac{e-1}{2}$.

Harder example Find the volume inside the sphere $\rho = 2$ and above the cone $\phi = \frac{\pi}{3}$.

Solution

$$\int_0^{2\pi} \int_0^{\pi/3} \int_0^2 \rho^2 \sin \phi d\rho d\phi d\theta = 2\pi \cdot (1 - \cos \frac{\pi}{3}) \cdot \frac{8}{3}.$$

⇒ $\frac{8\pi}{3}$.

Jacobian: over $0 \leq x + y \leq 1$, $0 \leq x - y \leq 2$, $\iint (x + y) dA = \int_0^2 \int_0^1 u \cdot \frac{1}{2} du dv = \frac{1}{2}$.

Traps Forgetting r or $\rho^2 \sin \phi$. When reversing, the new limits come from the sketch, not by swapping symbols.

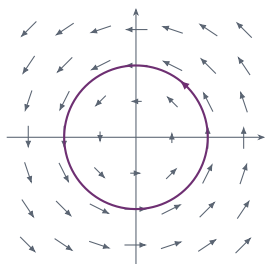
Method First ask: is $\mathbf{F}$ conservative? (Test $Q_x = P_y$, or $\nabla \times \mathbf{F} = \mathbf{0}$, on a simply connected domain.) If yes, find φ and evaluate at the endpoints; the path is irrelevant. If not: parametrise and integrate, or, for a closed plane curve, use Green's theorem to turn it into a double integral.

Key formulas

$$\begin{aligned}\int_C \mathbf{F} \cdot d\mathbf{r} &= \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt \\ \int_C f ds &= \int_a^b f(\mathbf{r}(t)) |\mathbf{r}'(t)| dt \\ \mathbf{F} = \nabla \varphi &\Rightarrow \int_C \mathbf{F} \cdot d\mathbf{r} = \varphi(B) - \varphi(A) \\ \oint_C P dx + Q dy &= \iint_R (Q_x - P_y) dA \\ \text{Area} &= \frac{1}{2} \oint x dy - y dx\end{aligned}$$

Problem type → first move

work along a path	conservative? then $\varphi(B) - \varphi(A)$
closed curve in the plane	Green (counter-clockwise)
find the potential	integrate P in x , match Q
mass of a wire	$\int \rho ds$
region with a hole	Green on both boundaries



$\mathbf{F} = (-y, x)$ circulates: $Q_x - P_y = 2$, so $\oint = 2 \times \text{area}$. Not conservative.

Example $\mathbf{F} = (2xy, x^2 + 3y^2)$. Find $\int_C \mathbf{F} \cdot d\mathbf{r}$ from $(0,0)$ to $(1,2)$ along any path.

Solution $P_y = 2x = Q_x$. $\varphi = x^2y + g(y)$, and $\varphi_y = x^2 + g'$ so $g = y^3$. $\varphi(1,2) - \varphi(0,0) = 2 + 8$.

⇒ 10.

Harder example Evaluate $\oint_C y^3 dx - x^3 dy$ around the unit circle, counter-clockwise.

Solution Green: $Q_x - P_y = -3x^2 - 3y^2$, so $\iint -3r^2 dA = -3 \int_0^{2\pi} \int_0^1 r^3 dr d\theta$.

⇒ $-\frac{3\pi}{2}$.

Traps Green needs the curve counter-clockwise and $\mathbf{F}$ smooth inside: $\frac{(-y,x)}{x^2+y^2}$ has $Q_x = P_y$ but circulation 2π round the origin.

Surface integrals, flux, Stokes' theorem and the divergence theorem

42

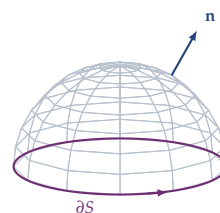
Method Flux through a closed surface: divergence theorem, almost always. Circulation round a curve, or flux of a curl: Stokes, and swap the surface for the simplest one with the same boundary (often a flat disc). Otherwise parametrise: $\mathbf{r}(u,v)$, normal $\mathbf{r}_u \times \mathbf{r}_v$, oriented as asked.

Key formulas

$$\begin{aligned}\iint_S f dS &= \iint_D f |\mathbf{r}_u \times \mathbf{r}_v| du dv \\ \iint_S \mathbf{F} \cdot d\mathbf{S} &= \iint_D \mathbf{F} \cdot (\mathbf{r}_u \times \mathbf{r}_v) du dv \\ z = g(x,y), \text{ upward: } d\mathbf{S} &= (-g_x, -g_y, 1) dA \\ \oint_{\partial S} \mathbf{F} \cdot d\mathbf{r} &= \iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} \\ \iint_{\partial E \text{ closed}} \mathbf{F} \cdot d\mathbf{S} &= \iiint_E \nabla \cdot \mathbf{F} dV\end{aligned}$$

Problem type → first move

flux out of a closed surface	$\iiint \nabla \cdot \mathbf{F} dV$
open surface, flux	close it, subtract the lid
flux of $\nabla \times \mathbf{F}$	Stokes: boundary, or a flatter surface
circulation round a space curve	Stokes, any surface it bounds
surface area	$\iint \mathbf{r}_u \times \mathbf{r}_v $



Upper hemisphere with outward normal; its boundary circle runs counter-clockwise seen from above.

Example Find the outward flux of $\mathbf{F} = (x, y, z)$ through the sphere of radius a .

Solution $\nabla \cdot \mathbf{F} = 3$, so the flux is $3 \cdot \frac{4}{3} \pi a^3$. Check: $\mathbf{F} \cdot \mathbf{n} = a$ on the sphere, times area $4\pi a^2$.

⇒ $4\pi a^3$.

Harder example $\mathbf{F} = (-y, x, z)$. Find $\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$ over the upper unit hemisphere, normal outward.

Solution $\nabla \times \mathbf{F} = (0, 0, 2)$. By Stokes, use the flat unit disc instead (same boundary, normal $\mathbf{k}$): $2 \times \pi$. Or $\oint -y dx + x dy = \int_0^{2\pi} 1 dt$ round the circle.

⇒ 2π .

Traps Orientation: the normal and the boundary direction must obey the right-hand rule. The divergence theorem needs a closed surface: an open one must be capped first.

Practice problems

TWO PER BLOCK · ANSWERS AT THE END

The tag after each number is the block of the guide that covers it. Try each one cold: name the problem type and the first move before you write anything, then check against the key.

Part I · technique selector

1. [I1] $\int_0^1 \frac{\ln(1+x)}{1+x^2} dx$
2. [I1] $\int_{-1}^1 (x^3 \cos x + x^2) dx$
3. [I2] $\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2}$
4. [I1] $\int_0^\infty x e^{-2x} dx$

Part II · calculus

5. [1] $\lim_{x \rightarrow 0} \frac{1 - \cos 3x}{x^2}$
6. [1] Show that $x^5 + x = 1$ has exactly one real root, and give it to two decimal places.
7. [2] Differentiate $\sqrt{x}$ from first principles.
8. [2] Is $f(x) = x|x|$ differentiable at 0? Find $f'(x)$.
9. [3] Find $\frac{dy}{dx}$ at $(1, 2)$ on $x^2 + xy + y^2 = 7$.
10. [3] Find $\frac{dy}{dx}$ at $(0, 1)$ on $e^{xy} = x + y$.
11. [4] Differentiate $\arcsin(2x)$.
12. [4] Find $\frac{d}{dx} x^x$ at $x = 1$.
13. [5] $x = \cos t$, $y = \sin 2t$. Find $\frac{dy}{dx}$ at $t = \frac{\pi}{6}$.
14. [5] Find the length of $x = t^2$, $y = \frac{2}{3}t^3$, $0 \leq t \leq 1$.
15. [6] Find the tangent to $y = \ln x$ that passes through the origin.
16. [6] Use a linear approximation to estimate $\sqrt{4.1}$.
17. [7] A sphere's radius grows at 2 cm/s. How fast is its volume growing when $r = 5$ cm?
18. [7] A cone (vertex down, radius 3 m, height 6 m) fills at $2 \text{ m}^3/\text{min}$. How fast is the depth rising when it is 4 m?
19. [8] Find the largest rectangle, sides parallel to the axes, inscribed in $\frac{x^2}{9} + \frac{y^2}{4} = 1$.
20. [8] Find the shortest distance from $(0, 2)$ to the parabola $y = x^2$.
21. [9] Find the extrema and inflection points of $y = xe^{-x}$.
22. [9] Find all asymptotes of $y = \frac{x^2 + 1}{x - 1}$.
23. [10] $v(t) = 3t^2 - 12t + 9$ on $[0, 4]$. Find the displacement and the total distance.
24. [10] For the same particle, find the acceleration each time it is at rest.
25. [11] $\int_0^1 \frac{x}{(1+x^2)^2} dx$
26. [11] $\int_1^e \frac{(\ln x)^2}{x} dx$
27. [12] $\int_0^1 x e^{2x} dx$

28. [12] $\int_0^1 \arctan x dx$
 29. [13] $\int_2^3 \frac{dx}{x^2 - 1}$
 30. [13] $\int_0^1 \frac{dx}{x^2 + 2x + 2}$
 31. [14] $\int_0^{\pi/2} \sin^3 x dx$
 32. [14] $\int_0^1 \frac{x^2}{\sqrt{4-x^2}} dx$
 33. [15] Find $\frac{d}{dx} \int_0^{x^3} e^{-t^2} dt$.
 34. [15] $g(x) = \int_1^{x^3} \sqrt{1+t} dt$. Find $g'(1)$.
 35. [16] Find the area between $y = \sin x$ and $y = \cos x$ for $0 \leq x \leq \frac{\pi}{2}$.
 36. [16] Find the total area enclosed by $y = x^3$ and $y = x$.
 37. [17] Rotate $y = \sqrt{x}$, $0 \leq x \leq 4$, about the x -axis. Find the volume.
 38. [17] Rotate the region between $y = x$ and $y = x^2$ about the y -axis. Find the volume.
 39. [18] $\int_1^\infty \frac{\ln x}{x^2} dx$
 40. [18] Find the length of $y = x^{3/2}$, $0 \leq x \leq 4$.
 41. [19] Solve $y' = 2xy$, $y(0) = 3$.
 42. [19] Solve $y' + \frac{y}{x} = x^2$, $y(1) = 1$.
 43. [20] $\sum_{n=1}^\infty \frac{1}{n(n+1)(n+2)}$
 44. [20] $\sum_{n=1}^\infty \frac{n^2}{2^n}$
 45. [21] Find the Maclaurin series of $e^x \sin x$ up to x^3 .
 46. [21] Find the radius of convergence of $\sum \frac{n! x^n}{n^n}$.
- Part III · algebra, trigonometry, geometry, discrete**
47. [22] Find the area of one petal of $r = \sin 3\theta$.
 48. [22] Write $r = 4 \sin \theta$ in Cartesian form and describe it.
 49. [23] $2x^3 + ax^2 + bx + 3$ has a factor $x + 1$ and leaves remainder 15 on division by $x - 2$. Find a and b .
 50. [23] α, β, γ are the roots of $x^3 + 3x - 1 = 0$. Find $\alpha^3 + \beta^3 + \gamma^3$.
 51. [24] Solve $\frac{x-1}{x+3} < 2$.
 52. [24] Simplify $\sqrt{11 - 6\sqrt{2}}$.
 53. [25] $f(x) = \sqrt{x-1}$, $g(x) = x^2$. Find the domain of $f \circ g$.
 54. [25] Find the inverse of $f(x) = e^{2x} + 1$, with its domain.
 55. [26] Solve $9^x - 4 \cdot 3^x + 3 = 0$.

56. [26] Solve $\log_3(x+1) + \log_3(x-1) = 1$.
57. [27] Solve $\tan^2 x = 3$ for $0 \leq x < \pi$.
58. [27] Solve $\cos 2x = \sin x$ for $0 \leq x < 2\pi$.
59. [28] Solve $\sqrt{3}\sin x + \cos x = 1$ for $0 \leq x < 2\pi$.
60. [28] Find $\cos 75^\circ$ exactly.
61. [29] A triangle has sides 5, 6, 7. Find its largest angle and its area.
62. [29] A box is $3 \times 4 \times 12$. Find the angle between a space diagonal and the base.
63. [30] Prove $6 \mid n^3 + 5n$ for all $n \geq 1$.
64. [30] Prove $n! > 2^n$ for all $n \geq 4$.
65. [31] Find the constant term of $(x^2 + \frac{1}{x})^6$.
66. [31] Find the coefficient of x^2 in $(1+2x)^5(1-x)^3$.
67. [32] Evaluate $(1 - i\sqrt{3})^6$.
68. [32] Solve $z^2 = 3 + 4i$.
69. [33] Evaluate $\det \begin{pmatrix} 2 & 0 & 1 \\ 1 & 3 & -1 \\ 0 & 1 & 4 \end{pmatrix}$.
70. [33] Invert $\begin{pmatrix} 4 & 7 \\ 2 & 4 \end{pmatrix}$.
71. [34] Solve $2x + y - z = 1$, $x - y + 2z = 5$, $3x + 2y + z = 10$.
72. [34] For which k does $kx + y = 1$, $x + ky = 1$ have no solution? Infinitely many?
73. [35] Find the angle between the planes $x + y + z = 1$ and $x - y + 2z = 3$.

74. [35] Where does $\mathbf{r} = (1, 2, 3) + t(1, 0, -1)$ meet the plane $x + 2y + z = 12$?
75. [36] A committee of 4 is chosen from 6 men and 5 women. How many have at least 2 women?
76. [36] Find the probability of at least one six in four rolls of a die.

Part IV · multivariable calculus

77. [37] Show that $f = \ln(x^2 + y^2)$ satisfies $f_{xx} + f_{yy} = 0$.
78. [37] $z = x^2y$, $x = \cos t$, $y = \sin t$. Find $\frac{dz}{dt}$ at $t = \frac{\pi}{4}$.
79. [38] $f = xe^{y^2}$. Find the rate of change at $(2, 0)$ in the direction of $(3, 4)$.
80. [38] Find the tangent plane to $z = x^2 + y^2$ at $(1, 2, 5)$.
81. [39] Find and classify the critical point of $f = x^2 + xy + y^2 - 3x$.
82. [39] Find the maximum of xy on the ellipse $x^2 + 4y^2 = 8$.
83. [40] $\iint_{x^2+y^2 \leq 4} \sqrt{x^2 + y^2} dA$
84. [40] $\int_0^1 \int_{\sqrt{y}}^1 \sqrt{x^3 + 1} dx dy$
85. [41] $\int_C (x + y) ds$, C the segment from $(0, 0)$ to $(3, 4)$.
86. [41] Find $\oint_C y dx - x dy$ round the unit circle, counter-clockwise.
87. [42] Find the outward flux of (x^2, y^2, z^2) through the surface of the unit cube $[0, 1]^3$.
88. [42] Find the area of the paraboloid $z = x^2 + y^2$ below $z = 2$.

Answers to the practice problems

FINAL ANSWERS ONLY

1. $\frac{\pi}{8} \ln 2$ (substitute $x = \tan \theta$, then King)
2. $\frac{2}{3}$ (odd part vanishes)
3. $\frac{1}{2}$
4. $\frac{1}{4}$
5. $\frac{9}{2}$
6. one root, since $f' = 5x^4 + 1 > 0$; $x \approx 0.75$
7. $\frac{1}{2\sqrt{x}}$
8. yes; $f'(x) = 2|x|$
9. $-\frac{4}{5}$
10. 0
11. $\frac{2}{\sqrt{1-4x^2}}$
12. 1
13. -2
14. $\frac{2}{3}(2\sqrt{2}-1)$
15. $y = \frac{x}{e}$, touching at $(e, 1)$
16. 2.025
17. $200\pi \text{ cm}^3/\text{s}$
18. $\frac{1}{2\pi} \text{ m/min}$
19. 12, at $(\frac{3}{\sqrt{2}}, \sqrt{2})$
20. $\frac{\sqrt{7}}{2}$
21. max $(1, \frac{1}{e})$; inflection $(2, \frac{2}{e^2})$
22. $x = 1$ and $y = x + 1$
23. displacement 4; distance 12
24. $t = 1$: $a = -6$; $t = 3$: $a = 6$
25. $\frac{1}{4}$
26. $\frac{1}{3}$
27. $\frac{e^2+1}{4}$
28. $\frac{\pi}{4} - \frac{1}{2} \ln 2$
29. $\frac{1}{2} \ln \frac{3}{2}$
30. $\arctan 2 - \frac{\pi}{4}$
31. $\frac{2}{3}$
32. $\frac{\pi}{3} - \frac{\sqrt{3}}{2}$
33. $2xe^{-x^4}$
34. $3\sqrt{2}$
35. $2\sqrt{2} - 2$
36. $\frac{1}{2}$
37. 8π
38. $\frac{\pi}{6}$
39. 1
40. $\frac{8}{27}(10\sqrt{10}-1)$
41. $y = 3e^{x^2}$
42. $y = \frac{x^3}{4} + \frac{3}{4x}$
43. $\frac{1}{4}$
44. 6
45. $x + x^2 + \frac{x^3}{3}$
46. e
47. $\frac{\pi}{12}$
48. $x^2 + (y-2)^2 = 4$: circle, centre $(0, 2)$, radius 2
49. $a = -1, b = 0$
50. 3
51. $x < -7$ or $x > -3$
52. $3 - \sqrt{2}$
53. $|x| \geq 1$
54. $\frac{1}{2} \ln(x-1), x > 1$
55. $x = 0$ or 1
56. $x = 2$ ($x = -2$ rejected)
57. $\frac{\pi}{3}, \frac{2\pi}{3}$
58. $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$
59. 0, $\frac{2\pi}{3}$
60. $\frac{\sqrt{6}-\sqrt{2}}{4}$
61. $\arccos \frac{1}{5} \approx 78.5^\circ$; area $6\sqrt{6}$
62. $\arctan \frac{12}{5} \approx 67.4^\circ$
63. step: $f(k+1) = f(k) + 3k(k+1) + 6$, and $k(k+1)$ is even
64. base $24 > 16$; step: $(k+1)! = (k+1)k! > 2 \cdot 2^k$
65. 15
66. 13
67. 64
68. $\pm(2+i)$
69. 27
70. $\frac{1}{2} \begin{pmatrix} 4 & -7 \\ -2 & 4 \end{pmatrix}$
71. $(1, 2, 3)$
72. none: $k = -1$; infinitely many: $k = 1$
73. $\arccos \frac{\sqrt{2}}{3} \approx 61.9^\circ$
74. nowhere: the line is parallel to the plane
75. 215
76. $\frac{671}{1296} \approx 0.518$
77. $f_{xx} = \frac{2(y^2-x^2)}{(x^2+y^2)^2} = -f_{yy}$
78. $-\frac{\sqrt{2}}{4}$
79. $\frac{11}{5}$
80. $z = 2x + 4y - 5$
81. $(2, -1)$, minimum, $f = -3$
82. 2, at $\pm(2, 1)$
83. $\frac{16\pi}{3}$
84. $\frac{2}{9}(2\sqrt{2}-1)$
85. $\frac{35}{2}$
86. -2π
87. 3
88. $\frac{13\pi}{3}$